

REVIEW

Discovering the unseen: A performance comparison of taxonomic classification methods for unknown DNA barcodes

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Handling Editor: Pablo Duchén**Abstract**

1. DNA barcoding and metabarcoding have emerged as cost-efficient, standardized methods for characterizing local biodiversity. Based on the sequencing of a small targeted gene fragment, it is theoretically possible to identify a wide diversity of taxa by comparing them with reference sequence databases. However, a key challenge for accurate taxonomic classification is the incompleteness of such databases, leading to most query sequences lacking species-level matches.
2. Where species-level matches are missing, it may be possible to classify query sequences to a higher taxonomic rank, such as genus or family, based on the similarity of related reference taxa. The challenge then lies in confidently recognizing whether the sequence belongs to an unobserved (here, 'novel') taxon on a given rank.
3. We evaluate the performance and utility of several methods for taxonomic classification. Methods were assessed based on the classification accuracy of both observed and novel taxa, accuracy of prediction confidence estimates and computational resource use. We focus on two widely studied cases: the COI barcode for arthropods, and the ITS barcode for fungi, with the latter representing an instance with substantially greater sequence length variation within classes. To benchmark the classification of novel taxa, we used curated datasets with partially distinct taxonomic distributions between the training and test sets. Novel taxa occurred at all evaluated taxonomic ranks, such as novel species in observed genera and novel genera in observed families. We further assessed the effect on performance when shifting from full-length barcodes to shorter sequences as generated through metabarcoding in the test dataset.

Johanna Orsholm and Alessandro Zito contributed equally to this work.

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4. This study sheds light on the strengths and limitations of different classification algorithms across taxonomic groups and barcode characteristics. It demonstrates the supreme performance of phylogenetic placement methods (e.g., EPA-ng) for classification of arthropod COI barcodes and composition-based classifiers (e.g., SINTAX, RDP-NBC, IDTAXA) for fungal ITS. Differences likely reflect barcode properties: COI is alignable and evolutionary constrained, favouring phylogenetic placement, whereas ITS is too variable for reliable alignment but rich in short-motif (k-mer) signal, favouring composition-based classifiers. Across most algorithms, shorter sub-regions performed comparably to full-length barcodes, supporting the use of short reads for high-throughput metabarcoding.

KEYWORDS

algorithmic benchmarking, DNA barcoding, novel taxa, species identification, taxonomic classification

1 | INTRODUCTION

During recent decades, DNA barcoding has emerged as a versatile method to characterize biological samples (Hebert et al., 2003). DNA barcoding relies on short, standardized gene segments to classify unknown specimens within a Linnaean taxonomy using previously assembled DNA reference databases. Such a procedure is especially effective when combined with high-throughput sequencing technologies, which enable both rapid parallel sequencing of a large number of individual specimens (Shokralla et al., 2014) and analysis of mixed samples in bulk, also known as DNA metabarcoding (Riaz et al., 2011). As a result, sequencing pipelines have become indispensable tools for large-scale biodiversity analysis and monitoring, enabling the identification of species across different taxa and allowing researchers to rapidly assess species diversity in various ecosystems (Gostel & Kress, 2022; Niskanen et al., 2023; Van Klink et al., 2022).

Early work confirming the utility of DNA barcodes for taxonomic classification of unknown specimens has typically focussed on cases where there are close matches for the query sequences in the reference database (Deshpande et al., 2016; Hebert et al., 2003; Janzen et al., 2005; Schoch et al., 2012). However, these ideal conditions are rare: many taxonomic branches from highly diverse groups are either undescribed or currently lack reference sequences, as evident from a striking difference between current estimates of global species richness and the actual sizes of annotated sequence databases. For example, insect species diversity is estimated at 2.6–7.2 million (Stork, 2018), while well-curated reference databases account for roughly 1.2 million species-level sequence clusters with only ≈283 thousand annotated with formal species names (Ratnasingham et al., 2024, statistics downloaded on 2025-08-15 from <https://www.boldsystems.org/>). As for fungi, estimates oscillate between 1 and 12 million species, but databases present ≈170 thousand clusters with only ≈53 thousand formally annotated (Abarenkov et al., 2025, statistics downloaded on 2025-08-15 from <https://unite.ut.ee/>). Additionally, sequencing efforts have shown a clear geographic bias

towards Europe and North America (Khomich et al., 2018; Leandro et al., 2024), leaving many of the most diverse regions of the world underrepresented.

Due to the limited coverage of diverse taxonomic groups and geographical regions, a central challenge in DNA barcoding is classifying specimens whose taxa are missing from the reference database. We refer to taxa absent from the reference database as novel taxa, in contrast to observed taxa that are represented in the database. In this context, novel taxa include both described taxa lacking reference sequences and entirely undescribed ones. Accurate identification of novel taxa is critical for robust ecological inference, especially when a large proportion of the diversity is unknown. A better understanding of the taxonomic distribution of unknown diversity can also serve to guide sampling efforts targeted at biodiversity discovery and reference database improvement.

Given the large proportion of novel taxa, a classification algorithm must in practice (1) predict whether a query sequence belongs to an observed species and, if not, (2) correctly assign the higher taxonomic ranks of novel taxa up to the last observed level (e.g. the family of novel genera, or the genus of novel species). Conservative confidence thresholds for accepting predictions can reduce overclassification, where sequences from novel taxa are incorrectly identified as belonging to observed taxa, but may simultaneously lead to high rates of under-classification, where sequences from observed taxa are incorrectly identified as belonging to novel taxa. Classification of novel taxa is further complicated by the lack of a consistent relationship between sequence similarity and the lowest common rank across taxonomic branches (Edgar, 2018). These complexities call for well-calibrated estimations of prediction uncertainty (Somervuo et al., 2017), allowing researchers to make informed decisions about retaining classifications and incorporating uncertainty in downstream analyses.

Beyond reference coverage, classifier performance also depends on the biological properties of the barcode regions, which vary substantially among taxa. In principle, a genetic marker is suitable for

taxonomic classification if the variation between species is larger than the variation within species (Hebert et al., 2004; Somervu et al., 2017). These properties are seen in the mitochondrial cytochrome c oxidase subunit 1 (COI; Hebert et al., 2003) in animals, and in the nuclear ribosomal internal transcribed spacer (ITS; Schoch et al., 2012) in fungi. The traditional COI barcode region, known as the Folmer region, is usually 658 base pairs (bp) long (Folmer et al., 1994), and allows for relatively simple sequence alignments thanks to the strong functional constraints on its length and amino acid sequence (Pentinsaari et al., 2016). In contrast, the ITS barcode consists of two spacers—ITS1 and ITS2—surrounding the non-coding 5.8S rRNA, each varying widely in both base arrangement and sequence length (Schoch et al., 2012). As a consequence, it is practically impossible to produce reliable multiple sequence alignments across broad taxonomic groups (Lindahl et al., 2013). In fungi, ITS varies between 400 and 800bp (Manter & Vivanco, 2007), but in some cases, it may be up to 1600bp in length (Feibelman et al., 1994). Hence, COI and ITS represent opposite ends within the range of characteristics among commonly used barcodes (Guo et al., 2022) and therefore pose distinct challenges for taxonomic classification.

1.1 | Overview of taxonomic classifiers

Over the past two decades, a wealth of algorithms have been developed to classify query sequences using taxonomically labelled reference sequences, often employing markedly different strategies. However, no classification algorithm is universally superior to the others a priori. Rather, each method is originally designed for a specific purpose, either favouring speed of prediction, unbiased estimation of prediction confidence estimates, or the capacity to recognize novel taxa. Hleap et al. (2021) grouped taxonomic classification algorithms into four categories: (1) similarity-based; (2) composition-based; (3) probabilistic; and (4) phylogenetic. We argue that recent advances in machine learning and deep learning methodologies have since led to the birth of a novel fifth category, that of (5) *neural network classifiers*. From each category, we selected representative methods and tested their performance; See Table 1 for a summary, and refer to Table S1 for additional algorithms and motivations for their exclusion from the comparison.

Similarity-based classifiers use pairwise alignment of query sequences to the reference sequences as the basis of classification. BLAST (Altschul et al., 1990; Camacho et al., 2009) is a commonly used tool for sequence similarity search. While it is not specifically designed for taxonomic classification, a widespread first approach is to simply take the taxonomy of the top BLAST hit; see Bonato et al. (2022) and Vass et al. (2022) for examples. A slightly more sophisticated approach uses a priori similarity thresholds to establish whether to accept the top hit classification at each taxonomic rank (Tedesoo, Albertsen, et al., 2021). Here, a similarity threshold of 97% has been widely used for species-level identification across different taxa and barcode markers (Porter & Hajibabaei, 2020; Tedesoo et al., 2022). These thresholds can be further optimized for different

TABLE 1 Features of selected taxonomic classification algorithms.

Algorithm	Data type	Amino	Stopping	Novelty	Confidence	Outside info	Implementation language	Interface
BLAST top-hit	Similarity	✓	✓	✓	✓		C++	cmd, web interface
BLAST threshold	Similarity	✓	✓	✓	✓	Thresholds	C++, awk script	cmd
CREST4	Similarity	✓	✓	✓	✓		Python + BLAST	Python pkg., cmd
dnabarcoder	Similarity	✓	✓	✓	✓		Python + BLAST	cmd
RDP-NBC	k-mer profile	✓	✓	✓	✓		Java	cmd
SINTAX	k-mer profile	✓	✓	✓	✓		C++	cmd
IDTAXA	k-mer profile	✓	✓	✓	✓		C, R	R pkg.
ProtaxA	Similarity from alignment	✓	✓	✓	✓	Full taxonomy	C, Perl, R (training)	cmd, R script
Protax	Similarity	✓	✓	✓	✓	Full taxonomy	Perl + USEARCH, R (training)	cmd, R script
EPA-ng	Alignment	✓	✓	✓	✓	Backbone tree	C++ (EPA-ng; Gappa; IQ-TREE2)	cmd
BayesANT	Alignment or k-mer profile	✓	✓	✓	✓		R	R pkg.
MycoAI CNN	k-mer profile	✓	✓	✓	✓		Python (PyTorch)	Python mod., cmd
MycoAI BERT	BPE	✓	✓	✓	✓		Python (PyTorch)	Python mod., cmd

Note: Amino: All selected algorithms operate on nucleotide sequences; some can additionally be applied to amino acid sequences. Stopping: Does the algorithm have any mechanism to avoid over-classifying sequences? Novelty: Does the algorithm distinguish between low classification certainty for existing categories and prediction of new categories, which were not present in the reference? Confidence: Does the algorithm produce a numerical estimate on a fixed scale (i.e., 0–1 or 0–100) for the confidence of its classifications at each rank? Outside info: Does the algorithm allow additional data beyond the sequences and taxonomic labels of the training data? If so, what? Implementation language also lists any software dependencies.

taxa, as in dnabarcoder (Vu et al., 2022), or replaced by instead considering multiple hits simultaneously, assigning the last common ancestor (LCA) of the k nearest neighbours (KNN) from a BLAST search, as in CREST4 (Lanzén et al., 2012). Thresholding and KNN-consensus both limit over-classification, and the KNN-consensus method can further prevent over-classification in cases where closely related species share identical barcode sequences. However, neither method provides a numerical confidence estimate or distinguishes between uncertain classifications and novel taxa.

Composition-based classifiers avoid the need for computationally expensive pairwise alignments by decomposing both queries and reference sequences into k -mers, that is distinct overlapping subsequences of length k . A sequence can then be described by its k -mer profile—a fixed-length vector with one entry for each possible k -mer. The length of the vector is 4^k for nucleotide sequences and 20^k for amino acid sequences. From this category, we selected the RDP naïve Bayesian classifier (RDP-NBC, Wang et al., 2007), SINTAX (Edgar, 2016), and IDTAXA (Murali et al., 2018). All three models use bootstrap samples of the k -mer composition of the query sequences to produce numerical confidence estimates for the classification at each taxonomic rank. This helps control over-classification in a more granular way than the similarity search methods, but still cannot distinguish between uncertain classifications and novel taxa.

Probabilistic classifiers use a statistical framework to estimate the probability that a query sequence belongs to a given taxon at each taxonomic rank. From this category, we selected BayesANT (Zito et al., 2023) and PROTAX (Somervuo et al., 2016). BayesANT leverages Bayesian nonparametric species sampling priors (De Blasi et al., 2015; Pitman, 1996) to specifically model the taxonomic tree (Rigon et al., 2025). PROTAX is a model framework that can incorporate any covariates as predictors of taxonomy (e.g. sequence similarity, outputs from other classifiers), allowing users to tailor the model to better fit their specific needs. There are also concrete implementations designed and trained for specific taxonomic groups, such as FinPROTAX for arthropods (Roslin et al., 2022) and Protax-Fungi for fungi (Abarenkov et al., 2018), which can be readily applied without further model customization. Both BayesANT and PROTAX explicitly account for the possibility that a query sequence originated from a novel taxon and thus distinguish between classification uncertainty and novelty. Both have variants that can operate either on unaligned or globally aligned sequences, and while the latter may be both faster and higher performing, it is limited to sequences with relatively conserved length, such as the COI barcode for animals. BayesANT and PROTAX provide confidence estimates designed to be interpreted as well-calibrated probabilities and PROTAX can leverage information from a reference taxonomy that includes taxa lacking associated reference sequences.

Phylogenetic placement classifiers place query sequences individually into an existing phylogenetic tree generated from a set of reference sequences. In this framework, novel taxa are cleanly represented as placement on branches outside the clades which represent observed taxa at the rank under consideration. Construction of

the reference tree is left to the user, with a wide range of algorithms available but differing in speed and accuracy. There are several opportunities to incorporate additional information beyond the reference sequences at the stage of building the reference tree, including constraining the topology to conform to the taxonomic hierarchy or to published phylogenies. Alternatively, the reference tree can be generated using longer sequences or additional genes beyond the region covered by the query sequences. From this category we selected EPA-ng (Barbera et al., 2019), a maximum-likelihood based phylogenetic placement algorithm that calculates the likelihood of placement on each branch of the tree for each query sequence, according to a chosen evolutionary model. From these placements, Gappa (Czech et al., 2020), a helper tool for phylogenetic placement, can perform the mapping to taxonomic classification. Gappa can also report classification probabilities when they are provided by the phylogenetic placement algorithm. However, phylogenetic placement typically requires that the query and reference sequences are globally aligned, making it unsuitable for ITS fungal sequences. Therefore, we only tested this class of models for COI.

Neural network classifiers are united by their use of learned vector representations for barcode sequences, which allow them to automatically learn complex patterns from the reference data. They often generalize well within the represented groups, but may be over-confident when queries come from missing taxa. From this category, we selected MycoAI (Romeijn et al., 2024), which implements both a convolutional neural network (CNN) operating on sequences represented as k -mer profiles, and a bidirectional transformer (BERT) operating on byte-pair encoded (BPE) sequences, which represent the data using recurring variable-length motifs learned from the training data. Both architectures use hierarchical label smoothing during training and a multi-head output for the different taxonomic ranks in order to borrow information between taxonomic ranks and provide separate predictions and confidence estimates at each rank. Like the composition-based classifiers, MycoAI produces confidence estimates that can be used to reduce over-classification, but does not explicitly distinguish between prediction uncertainty and novel taxa.

Previous comparisons of taxonomic classifiers have found that simple top-hit algorithms, such as BLAST, often achieve the highest classification accuracy of observed species (Edgar, 2018; Hleap et al., 2021), making them attractive choices for study systems with extensive reference databases. However, classification accuracy typically declines with a decrease in the similarity between the query sequence and the top hit reference sequence (Edgar, 2018; Richardson et al., 2017), making it challenging to reliably classify novel taxa. The aim of this study was to evaluate the performance of taxonomic classification methods for DNA barcodes, specifically focussing on the prevalent issue of missing taxa in reference databases (Figure 1).

This study is the first to utilize real-world geographic and methodological splits to generate partially overlapping sequence sets, distinguishing our approach from previous benchmarks that relied on artificial thresholds (Edgar, 2018) or random in silico database partitioning (Richardson et al., 2017) to mimic taxonomic novelty. We

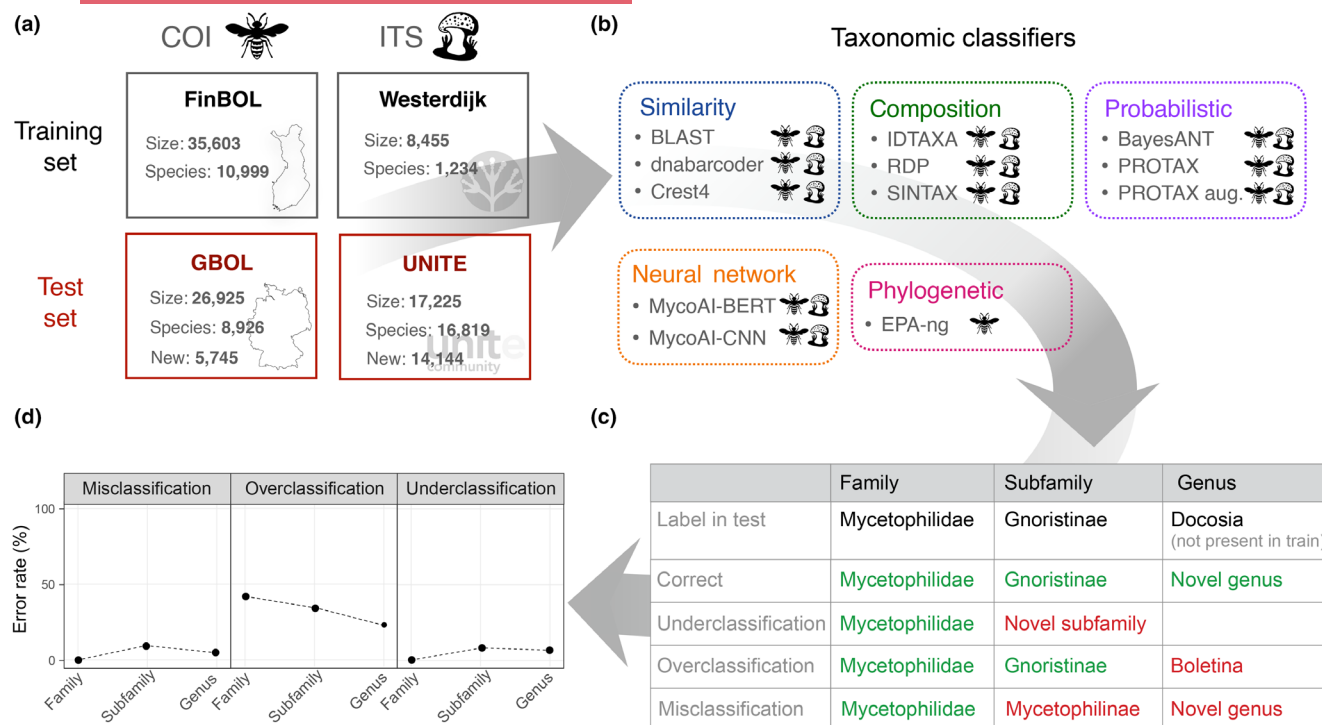


FIGURE 1 Overview of the analysis. (a) Datasets used in the analysis and the partial overlap between training and testing. (b) Main taxonomic classifiers and their categorization. We use each algorithm to classify all sequences from the test set, using the training set as reference data. Insect and mushroom icons indicate if the algorithm was evaluated on COI, ITS, or both. (c) Examples of classifier outputs representing different error types during the classification phase. Comparisons are made against the true labels in the test set. (d) Example of summarized classification results across taxonomic ranks.

argue that this natural distribution shift reflects the common reality of regional faunal differences and research biases that exist between available reference data and query sequences from diverse environmental samples. Additionally, we define novelty by taxonomic rank rather than similarity thresholds, making our results directly interpretable within standard taxonomy. Within this framework, we consider two distinct biological cases: the protein-coding COI barcode with conserved length and the non-coding ITS barcode of variable length. We evaluate performance on both full-length barcodes and shorter sub-regions of the query sequences. This comparison is motivated by practical constraints: due to the limitations on sequence length posed by some high-throughput sequencing methods, a shorter subregion of the markers is often targeted for DNA metabarcoding, resulting in around 400bp for COI (Elbrecht et al., 2019) and 300–400bp for ITS (Winand et al., 2025). Hence, our goal is to mimic the shorter metabarcoding reads, which are associated with lower sequencing costs but potentially at the expense of taxonomic resolution for novel lineages. We evaluate stand-alone programs representing five classifier categories, and we refrain from case-specific parameter optimization to reflect typical user scenarios. We assess these tools based on classification accuracy, calibration of confidence estimates, prediction coverage and computational resource use. A priori, we expect (i) a trade-off between accuracy and coverage, and (ii) higher accuracy at the expense of higher computational costs for algorithms that rely on complex models of taxonomic structure in sequence data.

2 | MATERIALS AND METHODS

2.1 | Data description and pre-processing

In our evaluation, models were trained on a set of barcodes, called the *training set*, which had a partial taxonomic overlap with the sequences used to assess model performance, the *test set*. We also generated a *testshort* set in which a standardized subregion of the sequences in the test set was selected to represent the shorter sequences that are often obtained with metabarcoding.

All sequences in both sets were associated with a species-level identification. However, in some cases higher ranks are missing from the classifications, which can occur for two reasons. First, the 'secondary ranks'—in our COI dataset subfamily and tribe—are considered optional, and are only used in diverse groups where additional levels of classification are desirable. Of our datasets, this only occurs in arthropod COI, because no secondary ranks are used in the fungal ITS data. Second, there may be genuine uncertainty about what higher taxa a species should belong to, such that taxonomists have chosen not to classify at all ranks. This situation is more common in our fungal ITS dataset. While the taxonomic annotations could contain some errors, for the purposes of this evaluation we treat them as the ground truth. To reflect realistic differences between training and test sets, we used sequences from different countries for COI and sequences derived from different methodological approaches for ITS.

2.1.1 | COI sequences: FinBOL And GBOL

For COI, we used DNA sequences from the Finnish Barcode of Life (FinBOL; Roslin et al., 2022) as the training set and from the German Barcode of Life (GBOL; German Barcode of Life Consortium, 2011) as the test set. FinBOL and GBOL aim to create comprehensive DNA barcode reference databases for multicellular organisms in Finland and Germany, respectively, using targeted sequencing of expert-identified specimens. Because they focus on separate regional faunas and, to some degree, reflect the research priorities of their respective taxonomic communities, the sets have only a partial taxonomic overlap.

For FinBOL, sequences and taxonomic annotations (including ranks class, order, family, subfamily, tribe, genus and species) were retrieved from the Barcode of Life Data Systems (BOLD; Ratnasingham & Hebert, 2007; Ratnasingham et al., 2024) data package 29-Mar-2024 (BOLD Systems, 2024) using the project code 'DS-FINPRO'. For GBOL, sequences were retrieved from the GBOL web portal (German Barcode of Life Consortium, 2011) using the filters 'Sub-/Phylum: Arthropoda', 'Collected: 11' and 'With Barcode: 11' to retrieve COI barcodes from a maximum of 11 specimens per species within Arthropoda. To avoid discrepancies between the taxonomy of the training and test sets, the Process-ID corresponding to records in BOLD was used to retrieve taxonomic information for each sequence in the test set.

Secondary taxonomic ranks, here subfamily and tribe, are not assigned to all taxa in the arthropod taxonomy. To represent taxa without secondary ranks, we used *dummy* as placeholder taxa. In our evaluation of COI classifications, we treated these placeholder taxa as equivalent: for example, the genera *Acanthosoma* and *Elasmotherus* were classified in the subfamily *Acanthosomatinae*, but did not have a tribe classification. At the tribe rank, they were therefore both represented as *Acanthosomatinae_dummy_tribe*. We avoided the use of the more standard taxonomic term '*incertae sedis*', meaning 'of uncertain placement', because the MycoAI classifier ignores taxa which end with this term, leading to unexpectedly poor performance in our tests at these ranks.

Several classifiers we tested, including dnabarcoder, SINTAX, and MycoAI, require explicitly named primary Linnean ranks. For these algorithms, we recoded the ranks above genus level as tribe → family, subfamily → order, family → class, order → phylum, and class → kingdom.

For both training and test sets, we retained only records that (1) were identified to the species level; (2) did not belong to a 'placeholder' taxon at the species level; (3) had been annotated as being a sequence of the COI-5P barcode region; and (4) were at least 600bp in length. Training and test sequences were aligned to the Folmer barcode region (Folmer et al., 1994), and amino acid translations were generated using MACSE v2.07 (Ranwez et al., 2018) and the COI coding sequence from *Drosophila melanogaster* (NCBI GenBank accession NC_024511.2, positions 1474..3009) as a reference. Sequences with less than 600 bases in the nucleotide alignment after removal of gap columns were removed from both nucleotide and amino acid

alignments, resulting in 35,603 training and 26,925 test sequences; see [Supporting Information S1](#) for further details. The final COI test set included novel taxa at all ranks from class to species, as displayed in [Figure 2](#).

We generated the testshort data set for COI by replacing the first 240 positions of the nucleotide test alignment and the first 80 positions of the amino acid test alignment (i.e. up to and including the BF3 primer binding site; Elbrecht et al., 2019) with gaps ('-'). Trimmed but unaligned training, test, and testshort sequences were generated by removing all gap characters from the aligned sequences.

2.1.2 | ITS sequences: Westerdijk culture collection and UNITE reference sequences

For ITS, we used DNA sequences from the Westerdijk Fungal Biodiversity Institute culture collection of yeasts (Vu et al., 2016) and filamentous fungi (Vu et al., 2019) as the training set. Our test set consisted of curated reference sequences from the UNITE fungal ITS database (Abarenkov et al., 2023). These are sequences that have been selected manually to represent species hypotheses. The Westerdijk Institute maintains the largest culture collection of living fungi in the world, while the UNITE database is a community-curated database of eukaryotic ITS sequences generated from diverse substrates and methods. The implied differences in scope and methodological breadth result in a partial taxonomic overlap.

For the training set, sequences were retrieved from National Center for Biotechnology Information (NCBI) Bioprojects PRJNA351778 (<https://www.ncbi.nlm.nih.gov/bioproject/351778>) and PRJNA422523 (<https://www.ncbi.nlm.nih.gov/bioproject/422523>) and matched with taxonomic annotations (including ranks kingdom, phylum, class, order, family, genus, and species) from the UNITE database (Abarenkov et al., 2023). For the test set, we downloaded the UNITE 10.0 general FASTA release from <https://doi.org/10.15156/BIO/2959332> and selected sequences annotated as reference sequences ('Refs'). Many sequences from the Westerdijk culture collection also occur as reference sequences in UNITE. To avoid inclusion of the same record in the train and test sets, we removed duplicate records from the training set if multiple sequences were available for that species hypothesis. If the duplicate sequence was the unique representative of a species hypothesis, we instead removed it from the test set.

We obtained taxonomic information for UNITE species hypotheses from Abarenkov (2024). For fungal species that do not include all primary taxonomic ranks in their classification, the UNITE database uses *incertae sedis* names as placeholder taxa, paired with the name of the closest enclosing taxon and its rank. For example, the genus *Ceratocladium* is classified in phylum *Ascomycota*, but not in any of the many described classes, orders or families within *Ascomycota*, so its family is listed in UNITE as *Ascomycota_fam_Incertae_sedis*. Multiple taxa with the same placeholder cannot be assumed equivalent; for instance, genus *Septogloeum* is also listed at the family level as *Ascomycota_fam_Incertae_sedis*, but this does not indicate that the genera *Ceratocladium* and *Septogloeum* belong

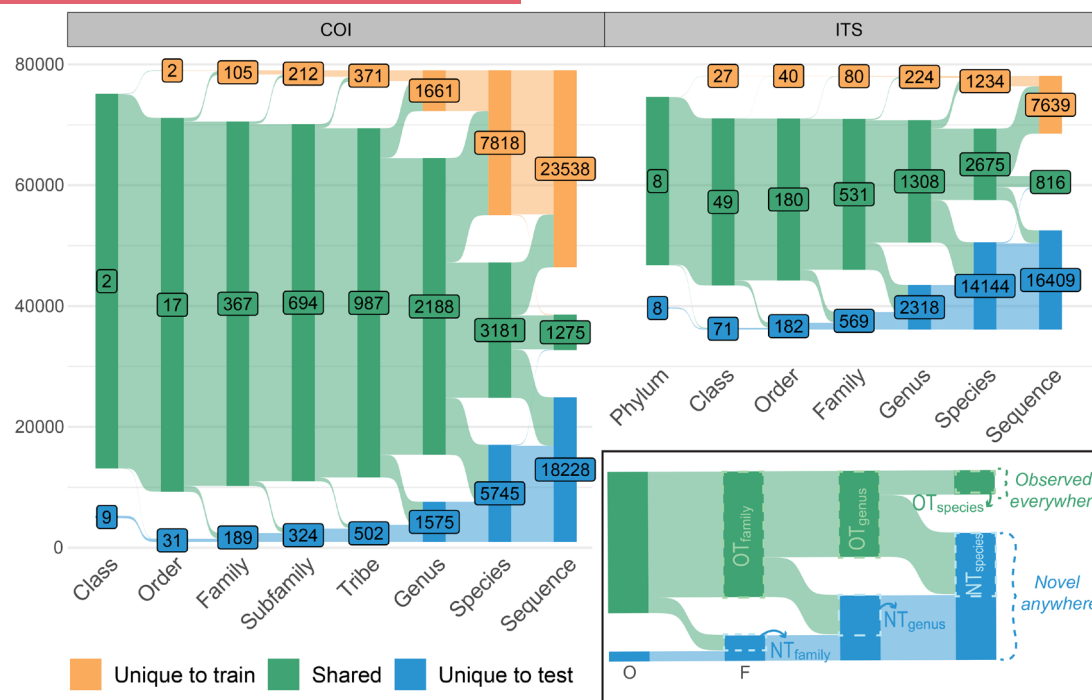


FIGURE 2 Taxonomic overlap between training set and test set for COI and ITS sequences. The x-axis displays the taxonomic ranks, with an additional column for unique DNA sequence variants. The height of columns and ribbons correlates to the number of sequences belonging to either taxa unique to the training set (orange), shared taxa (green) or taxa unique to the test set (blue). Numbers in squares show the count of unique taxa (or, for the last column, sequences) at each rank and group. The inset demonstrates how we partitioned the test data into *observed everywhere* and *novel anywhere*, as well as *observed taxa* (OT) and *novel taxa* (NT) at each rank, using ITS test sequences across ranks order to species as an example.

to the same natural group at the family level. To disambiguate such cases, we appended the nearest named lower rank to each *Incertae sedis* label, in the example case creating the family-level placeholders *Ascomycota_fam_Incertae_sedis_Ceratocladium* and *Ascomycota_fam_Incertae_sedis_Septogloeum*.

For both datasets, we extracted the ITS region with ITSx (Bengtsson-Palme et al., 2013), which uses profile hidden Markov models (HMMs) to detect the flanking small-subunit (SSU) and large-subunit (LSU) rRNA regions. When the included fragments of SSU or LSU are too short, ITSx sometimes fails to detect them. To solve the issue, we used cutadapt (Martin, 2011) to trim sequences matching the end of SSU, and LSUx (Furneaux et al., 2021) to detect and remove the beginning of LSU. Finally, we created testshort sequences including only ITS2 by deleting all nucleotide bases up to the end of 5.8S, as detected by the RNA families database (Rfam) covariance model RF00002 (Kalvari et al., 2021).

2.2 | Classification algorithms

We trained each classification algorithm on the training set and then classified all sequences in the test and testshort sets, as well as the amino acid versions of test and testshort for COI. Some methods feature many adjustable parameters for training and/or classification. To ensure a fair comparison, we used default settings when possible, or

parameter values that we believe reflect standard usage. Hence, we did not tune parameters to optimize results on our particular dataset, nor did we modify the algorithms from their most recent publicly available versions to improve their performance. Moreover, for algorithms that produce multiple alternative predictions per query at each rank, we selected the prediction with the highest probability or confidence (for brevity, hereafter collectively referred to as 'confidence') estimates, while remaining consistent with the predictions at higher ranks. We determined whether each prediction was correct by comparing it with the taxonomic annotation in the original test set. If the true taxon was missing from the training data, we considered the classification as correct if the taxon was predicted novel at the correct rank and all higher ranks were predicted correctly.

Some algorithms do not predict novelty but can refrain from making predictions for sequences with high classification uncertainty. Since a high proportion of missing predictions can inflate classification accuracy, we distinguish between results that allow missing predictions (MP) and those where predictions are enforced for all test sequences (FP). In the MP setting, we applied confidence thresholds for classification algorithms as recommended in the original publication and interpreted classifications below this threshold, as well as missing predictions, as neither correct nor incorrect. In the FP setting, we did not use any confidence threshold and interpreted missing predictions as the algorithm classifying the sequences as a novel taxon.

See Table 1 for an overview of the algorithms we tested, and Supporting Information S1.2 for execution details for each algorithm. Scripts used to install, train and test each algorithm are publicly available at the GitHub repository <https://github.com/jorsholm/taxclass>.

2.2.1 | Leveraging additional information

PROTAX and EPA-ng present opportunities for the user to augment the model with additional information. For PROTAX, we evaluated a base case, trained only on the training sequences, as well as an augmented case, where we supplied the full arthropod taxonomy from BOLD snapshot 29 March 2024 (BOLD Systems, 2024; Ratnasingham et al., 2024) for the COI case, and the full fungal taxonomy from the UNITE SH training data (Abarenkov, 2024) for the ITS case. For EPA-ng, we tested three different reference trees: one generated entirely de novo from the reference database with no constraints, one built with full taxonomic constraints, and one built with constraints based on previously established phylogenies for which relationships between families are resolved. We refer to the first case as the *free tree*, the second case as the *taxonomically constrained tree* ('tax'), and the third case as the *phylogenetically constrained tree* ('phyl').

2.3 | Data partitions

To explore the differences in algorithm performance on query sequences from observed or novel taxa, we used two complementary partitioning schemes: (1) a sequence-level split based on whether any rank was novel, and (2) a rank-specific split to evaluate performance separately at each taxonomic rank. In the sequence-level split, the *observed everywhere* partition contained test sequences whose taxa were observed at every rank considered, while the *novel anywhere* partition comprised test sequences that were novel at least one rank (Figure 2). In the rank-specific split, we split sequences into the *observed taxa* and the *novel taxa* partitions (Figure 2). The *observed taxa* partition at rank r contained test sequences whose taxon was observed at r , even if the sequence was novel at lower ranks. The *novel taxa* partition at rank r is instead made of test sequences whose taxon was novel at r and whose taxa at all higher ranks were observed (i.e. novelty first occurred at rank r). For example, a sequence from an observed order but a novel family was included in the *observed taxa* at the order rank and above, in *novel taxa* at the family rank, and was not included in either rank-specific partition for ranks below family. Consequently, the number of sequences in the rank-specific partitions differed across ranks.

2.4 | Performance metrics

To evaluate algorithm performance, we calculated classification accuracy where correct novelty prediction was labelled as a true positive. In other words, this metric was defined as follows:

$$\text{Accuracy} = \frac{TP}{N}$$

where TP was the number of correct predictions, including novel sequences correctly predicted as novel, and N was the number of sequences in the test set. For observed and novel taxa, we further calculated marginal and conditional recall. Recall is a performance metric used to measure of a model's ability to identify positive instances out of all true instances in the dataset (also known as sensitivity). Here, we defined *marginal recall* for each taxonomic rank as

$$\text{Marginal recall}_{t,r} = \frac{TP_{t,r}}{N_{t,r}}, \quad t \in \{\text{new, obs}\},$$

where $TP_{\text{new},r}$, $TP_{\text{obs},r}$ were the number of correct taxonomic predictions of sequences belonging to the groups of novel and observed taxa, respectively, at rank r , and $N_{\text{new},r}$, $N_{\text{obs},r}$ were the number of sequences in the test set belonging to the groups of novel and observed taxa at rank r , respectively. Note that $TP_{\text{new},r} + TP_{\text{obs},r} = TP$ and $N_{\text{new},r} + N_{\text{obs},r} = N$ at any rank. Marginal recall therefore measures accuracy at a given rank within a specific partition, enabling separate evaluation of algorithm performance at each taxonomic rank. For *conditional recall*, we considered only cases where the predictions for the higher ranks were correct. That is,

$$\text{Conditional recall}_{t,r} = \frac{TP_{t,r}}{TP_{t,r-1}} \quad t \in \{\text{new, obs}\},$$

where $TP_{t,r-1}$ is the number of sequences correctly predicted at rank $r-1$ for group t . Conditional recall therefore measures accuracy at a given rank within a specific partition, but only when the prediction was correct for higher ranks. It can therefore indicate how well a method refines classifications to deeper ranks once it has placed a sequence in the correct higher-level group.

For the full set of sequences, we calculated three further metrics of performance: (1) the *over-classification rate*, where sequences from novel taxa are incorrectly identified as belonging to observed taxa; (2) the *under-classification rate*, where sequences from observed taxa are incorrectly identified as belonging to novel taxa; and (3) the *misclassification rate*, where sequences belonging to one observed taxon are incorrectly identified as belonging to a different observed taxon, using the definitions by Edgar (2018). Error rates were evaluated under the MP setting, using recommended confidence thresholds when present.

We also calculated prediction coverage, defined as the proportion of sequences for which an algorithm assigned a taxonomic label, across all possible confidence thresholds. For algorithms that did not produce confidence estimates, coverage was calculated as a single value. We considered the MP setting, thus allowing missing predictions. However, for algorithms with a recommended confidence threshold to accept predictions, we computed prediction coverage across the full range of confidence thresholds.

Finally, we evaluated the calibration of confidence estimates by plotting the cumulative confidence of the predictions against the cumulative proportion of cases where the prediction was correct. For well-calibrated algorithms, confidence estimates and proportion of

correct predictions are equivalent: in cases where the confidence is 90%, the predictions are correct 90% of the time. Graphically, this results in a line close to 45 degrees. Instead, deviations from the 45-degree line are interpreted as under- or over-confidence. For algorithms missing confidence estimates, we assigned 100% confidence to all predictions.

2.5 | Computational resources

We executed algorithms on a compute node of the CSC Puhti supercomputer, a Linux high-performance computing cluster equipped with two Intel Xeon Gold 6230 CPUs for a total of 2×20 processing cores at 2.1 GHz. For each method, we ran the full training and classification task with allocations of 1, 4, 16 and 40 processor cores to test parallel scalability. If GPU execution was supported, we additionally ran methods in allocations with 1 CPU core and 1 Nvidia V100 GPU. We adjusted the memory and time allocations as needed for the algorithms to complete, with a maximum limit of 192 GiB of memory and 14 days of processing time. We measured elapsed time, maximum memory usage and average processor usage of the training and classification steps using the Gnu time utility. We also recorded disk space required to store the trained model.

3 | RESULTS

3.1 | Classification accuracy and recall

When classifying the *observed everywhere* partition under the FP setting, where we enforced predictions for all sequences, most algorithms exceeded 93% and 78% accuracy at the genus and species level, respectively, for both COI and ITS sequences (Figure 3a, for MP setting see Figure S3). For the *novel anywhere* partition, all algorithms exhibited a lower classification accuracy than for *observed everywhere*, ranging between 9% and 58% accuracy at the genus level for COI, and 12%–40% for ITS. Models that explicitly distinguished between classification uncertainty and novel taxon predictions, that is EPA-ng (COI only), BayesANT, and PROTAX, classified *novel anywhere* with accuracies between 12%–44% for COI, and 0%–29% for ITS. For COI, EPA-ng using taxonomically or phylogenetically constrained reference trees consistently achieved the highest classification accuracy across ranks. Phylogenetic placement on the free tree resulted in slightly lower accuracies at family level and below, but it was still among the most accurate algorithms. For ITS, composition-based classifiers exhibited the highest prediction accuracy for *novel anywhere* at genus level and above, with nearly identical accuracies for IDTAXA, RDP-NBC and SINTAX (Figure 3a). However, the inability of these classifiers to distinguish novelty from uncertainty led to 0% accuracy at species level for *novel anywhere* in the FP setting.

The *observed taxa* partition contained sequences which may be novel at lower ranks, and classification of these sequences may be

more difficult than those observed at species level. Accordingly, all algorithms showed lower marginal recall than accuracy at the corresponding rank (Figure 3b). This suggests that higher-level classifications were more error-prone when the sequence belonged to a novel taxon at a lower rank, compared to when a species-level match was available. Compared to the *observed everywhere* partition, the difference between algorithms was substantially larger for *observed taxa*. In general, similarity-based classifiers exhibited the lowest marginal recall for *observed taxa* for both COI and ITS, except BLAST top hit, which was consistently among the top-performing algorithms across all ranks for COI, and at ranks below order for ITS. For *novel taxa*, only EPA-ng, BayesANT and PROTAX explicitly predicted taxonomic novelty. Some similarity-based algorithms can abstain from making predictions, which we interpreted as novelty in the FP setting. Consequently, as expected, all composition-based and neural network classifiers exhibited zero marginal recall for *novel taxa*. Among the algorithms that explicitly predicted novelty, EPA-ng exhibited the overall highest marginal recall for COI, closely followed by BayesANT (Figure 3b). The class level was a notable exception, where PROTAX clearly outperformed both EPA-ng and BayesANT. For COI, BayesANT exhibited the highest conditional recall across ranks from family to species, suggesting it successfully predicted novelty in most cases where the higher ranks were predicted correctly (Figure S4). However, EPA-ng's slightly lower conditional recall may reflect that it correctly classified more sequences at higher ranks, leaving a larger set for evaluation at lower ranks and thus reducing the conditional proportion. For ITS, marginal recall for *novel taxa* was generally lower compared to COI (Figure 3b). However, at the genus and species levels, PROTAX achieved marginal recall comparable to its COI performance. At the genus level, the augmented PROTAX model performed markedly better than the base model. For ITS, BayesANT failed to correctly recover any novel taxa, in stark contrast to its COI performance. Consistent with this, BayesANT underestimated the number of sequences belonging to novel taxa across all ranks in ITS (Figure S7). By contrast, PROTAX overestimated the number of sequences belonging to novel taxa across all ranks. For COI, EPA-ng, BayesANT and PROTAX all overestimated the number of sequences belonging to novel taxa at the family level and below.

3.2 | Error rates

Most models showed very low misclassification rates (<5% for COI and <14% for ITS; Figure 4). Higher error rates were observed for MycoAI and BLAST top hit across both markers, and for BayesANT on ITS. In addition, at the ITS species level, RDP-NBC and SINTAX showed elevated error rates, misclassifying 20% and 14% of *observed everywhere*, respectively. Overall, over-classification and under-classification showed opposing patterns, highlighting the trade-off between the two. Even under the MP setting, there is no recommended confidence threshold or stopping rule for MycoAI or

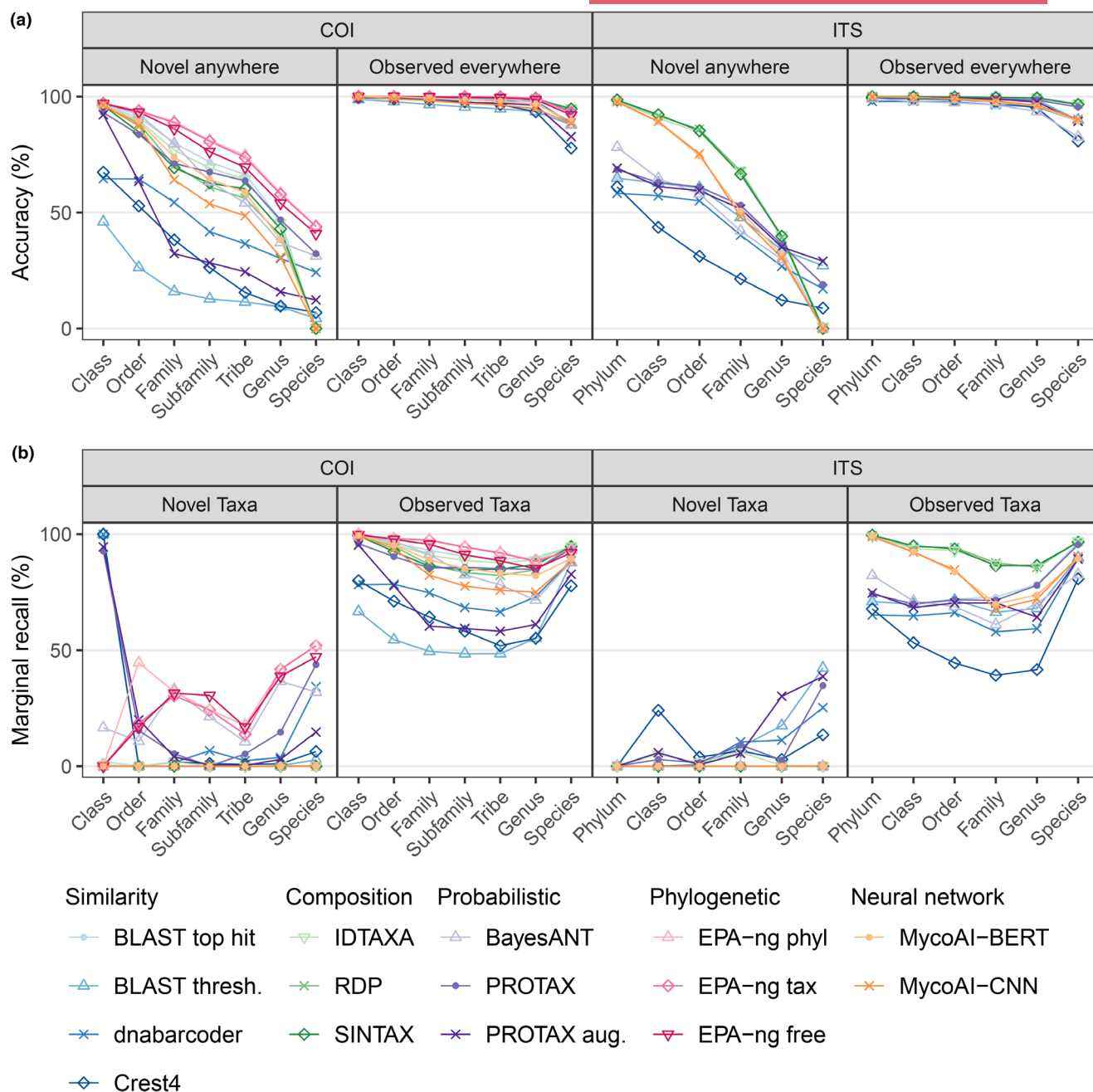


FIGURE 3 Accuracy and marginal recall of classification algorithms across taxonomic ranks under the FP setting, where predictions were enforced for all test sequences. (a) Shows accuracy, defined as the proportion of all taxonomic classifications that were correct for sequences in the *observed everywhere* and *novel anywhere* partitions. (b) Shows marginal recall, calculated as the proportion of correct predictions relative to the total number of sequences belonging to either *observed taxa* or *novel taxa* at that rank. Model categories are shown separately in Figure S2.

BLAST top hit, thus resulting in 100% over-classification for MycoAI for both markers, and nearly 100% over-classification for BLAST top hit and COI. For ITS, BLAST top hit had a higher proportion of missing predictions, resulting in lower over-classification rates. Among the algorithms that included a stopping rule, EPA-ng had the highest over-classification rate for COI. In contrast, BayesANT achieved both a low over-classification rate and a relatively low under-classification rate for COI. For ITS, however, BayesANT over-classified most novel taxa. RDP-NBC and SINTAX had overall low over-classification rates

for COI, except at the class level, where they over-classified 73% and 98% of novel taxa, respectively.

3.3 | Prediction coverage

There was a clear trade-off between accuracy and prediction coverage, where increasing confidence thresholds resulted in lower coverage but higher classification accuracy (Figure 5). The exact

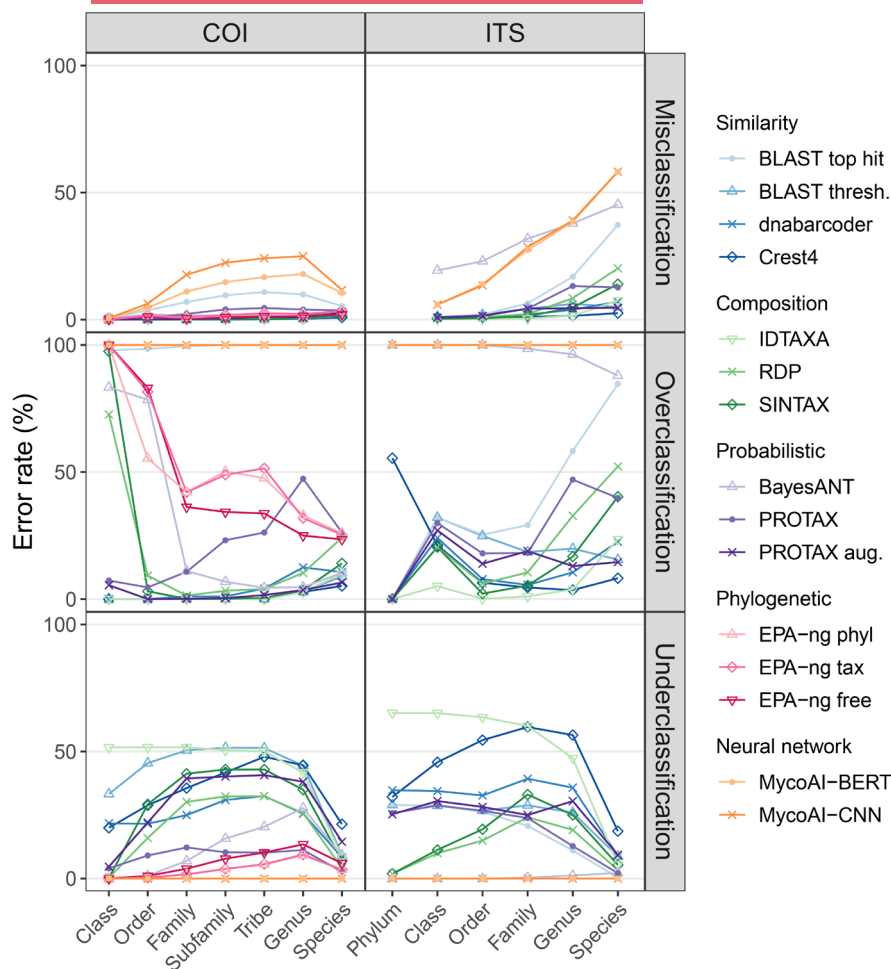


FIGURE 4 Rates of mis-, over- and under-classification of COI and ITS sequences across classification algorithms and taxonomic ranks. Error rates are shown under the MP setting, allowing missing predictions and implementing confidence thresholds for algorithms that recommend them. Model categories are shown separately in [Figure S5](#).

relationship between prediction coverage and accuracy varied between algorithms; some achieved moderate accuracies at high coverage, while others required aggressive filtering to achieve the same accuracy. Differences were especially pronounced for COI at the species level, where EPA-ng preserved high accuracy even at 100% prediction coverage. Algorithms without confidence estimates, such as BLAST top hit, appeared as fixed points in this trade-off space, classifying relatively few sequences but often with high correctness. Importantly, there was no consistent threshold that yielded a high degree of correct classifications across algorithms or genetic markers. The optimal operating point instead depends on whether accuracy or recall is more important for the application.

3.4 | Calibration of confidence estimates

For algorithms that produced confidence estimates, confidence was well-calibrated for the *observed everywhere* partition, with prediction confidence and prediction accuracy following a path close to the 45-degree line ([Figures S9](#) and [S10](#)). For the *novel anywhere* partition, MycoAI produced well-calibrated confidence estimates for both COI and ITS predictions. RDP-NBC, SINTAX, BayesANT, and PROTAX exhibited well-calibrated confidence estimates for

novel anywhere on COI, but overestimated confidence for ITS either across all ranks (BayesANT and PROTAX) or at lower ranks (RDP-NBC and SINTAX). The remaining algorithms exhibited either excessive confidence or a lack of it. In general, algorithms were more overconfident for ITS than for COI, especially at lower ranks ([Figures S9](#) and [S10](#)).

3.5 | Short and amino acid sequences

When evaluated on the testshort dataset, designed to mimic shorter metabarcoding reads, most algorithms classified *observed everywhere* with similar accuracies to full-length reads, with the exceptions of MycoAI for both markers and dnabarcoder and RDP-NBC for ITS ([Figure 6](#)). Notably, dnabarcoder assigned almost all ITS test-short sequences from *observed everywhere* to a novel class, yielding accuracy close to zero. For *novel anywhere*, classification accuracy declined on short reads compared to full-length reads, particularly for ITS at higher taxonomic ranks.

Translating COI to amino acid sequences yielded lower classification accuracies *observed everywhere* ([Figure S8](#)). However, BayesANT and Crest4 exhibited higher accuracies compared to nucleotide sequences for *novel anywhere*, especially at higher ranks.

FIGURE 5 Classification accuracy as a function of prediction coverage for different algorithms. The x-axis shows the proportion of sequences classified and the y-axis shows the proportion of correct predictions. Continuous lines indicate algorithms with confidence estimates, where varying the threshold yields a trade-off between coverage and accuracy. Single points represent algorithms without confidence estimates, corresponding to a fixed trade-off. We allowed missing predictions, but did not implement confidence thresholds for algorithms that recommend them. However, recommended thresholds are indicated by points along the continuous line. Model categories are shown separately in Figure S6.

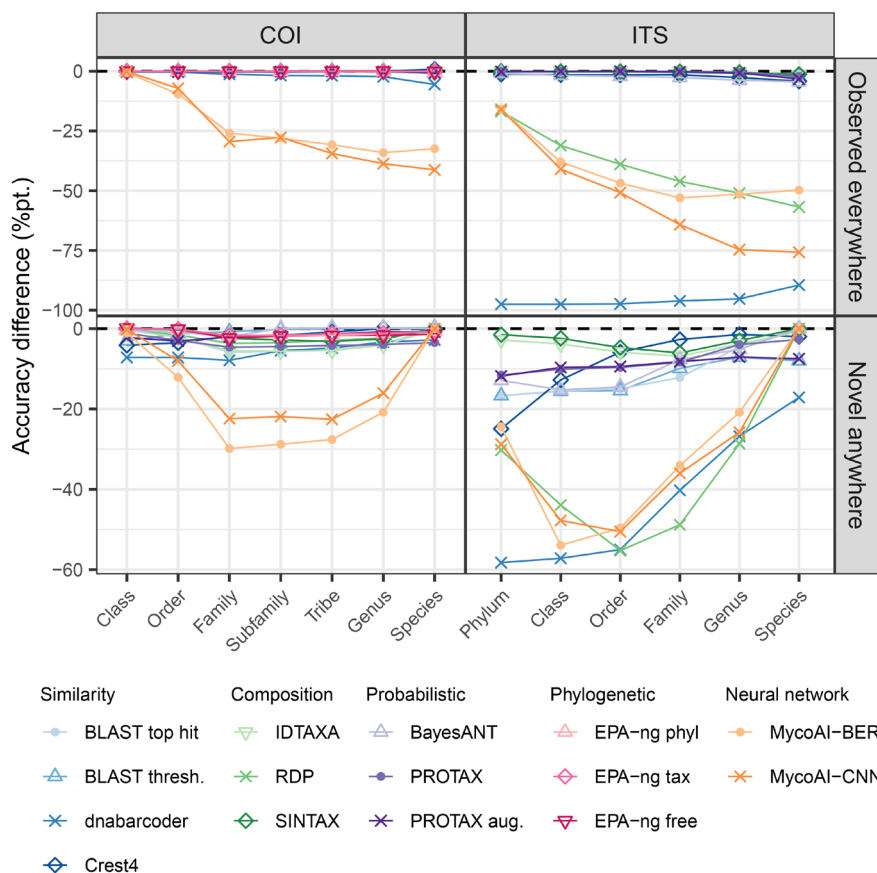
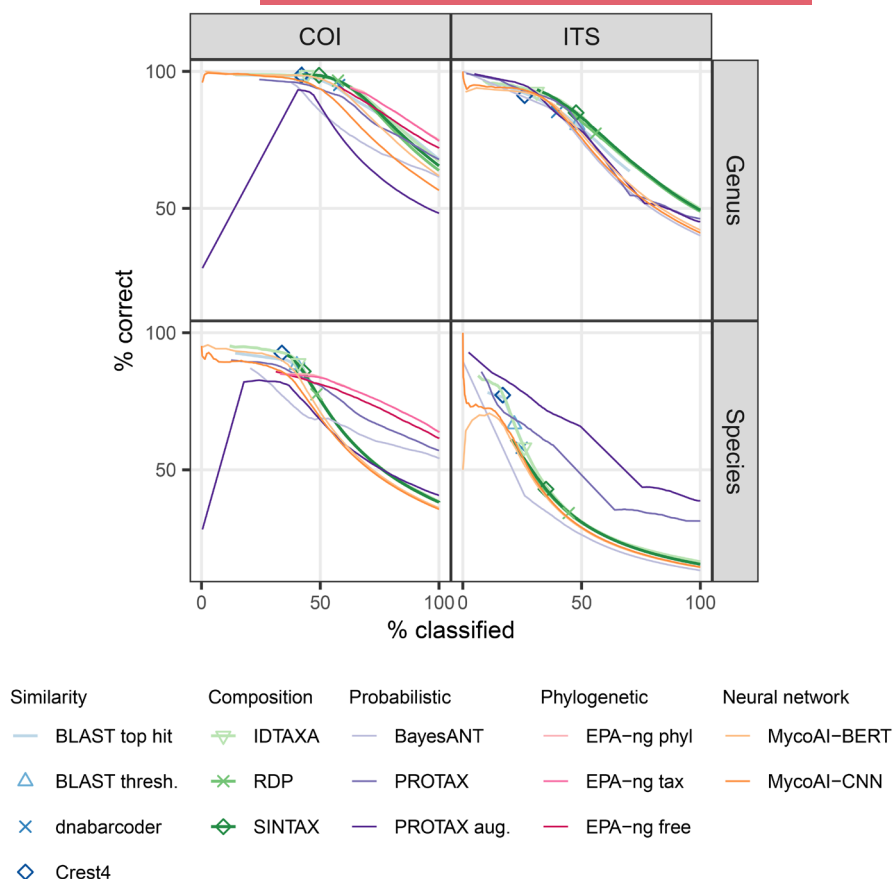


FIGURE 6 Change in classification accuracy when using full-length versus short sub-regions of the DNA barcodes, shown in percentage points. Line of zero difference is shown with a black dashed line. Negative values indicate lower accuracy of short reads than full-length barcodes, and vice versa.

3.6 | Computational resources

There was no clear relationship between classification performance and computational resource use. We show runtime in Figure 7 and report the full set of metrics in Table S3. Lightweight methods such as SINTAX achieved high accuracy with minimal runtimes and were the fastest overall when accounting for both training and classification, while resource-intensive algorithms such as MycoAI required substantially more time and memory for training without yielding superior accuracy. Once trained, however, classification with MycoAI was quick, with the CNN model having the shortest classification runtime overall. EPA-ng was substantially slower on amino acid sequences than nucleotide sequences due to the increased complexity of the substitution model. Most other algorithms were faster on translated sequences, likely due to their reduced length. Access to multiple CPU cores or a GPU reduced runtimes for most algorithms, especially during the classification phase, whereas fewer algorithms benefited substantially from parallelization during initialization or training, with MycoAI being a notable exception. Since training is typically a one-time cost, scalability during classification may be the more critical factor for real-world applications. Model size on disc also varied widely, with some MycoAI and BayesANT models occupying 200–400 MB compared to less than 30 MB for many similarity-, composition- or phylogenetic placement-based classifiers, which may make the latter more practical for deployment in large-scale or resource-limited settings.

4 | DISCUSSION

DNA barcoding has proved a versatile tool for biodiversity research, and its applications are continuously expanding and developing following advances in sequencing technologies (Gostel & Kress, 2022; Niskanen et al., 2023; Van Klink et al., 2022). For many purposes, however, reliable taxonomic assignment of the resulting barcode sequences is paramount for downstream analyses. The primary factor determining classification accuracy is the completeness of the reference database, as emphasized in previous studies (Edgar, 2018; Sickel et al., 2015; Taberlet et al., 2012) and strongly supported by our result in Figure 3. However, we have demonstrated that the sensitivity to incomplete reference data varies between taxonomic classifiers and that differences in performance between algorithms are more pronounced when species-level matches are lacking. Our results highlight that taxonomic classifiers vary in how they balance different types of classification errors, emphasizing the importance of careful selection of algorithms suitable for both the genetic marker targeted and the research question at hand.

4.1 | High classification accuracy comes at the cost of prediction coverage

We found a general *trade-off* between classification accuracy and prediction coverage across all methods (Figure 5), similar to previous studies (Edgar, 2018; Richardson et al., 2017). This outcome is intuitive:

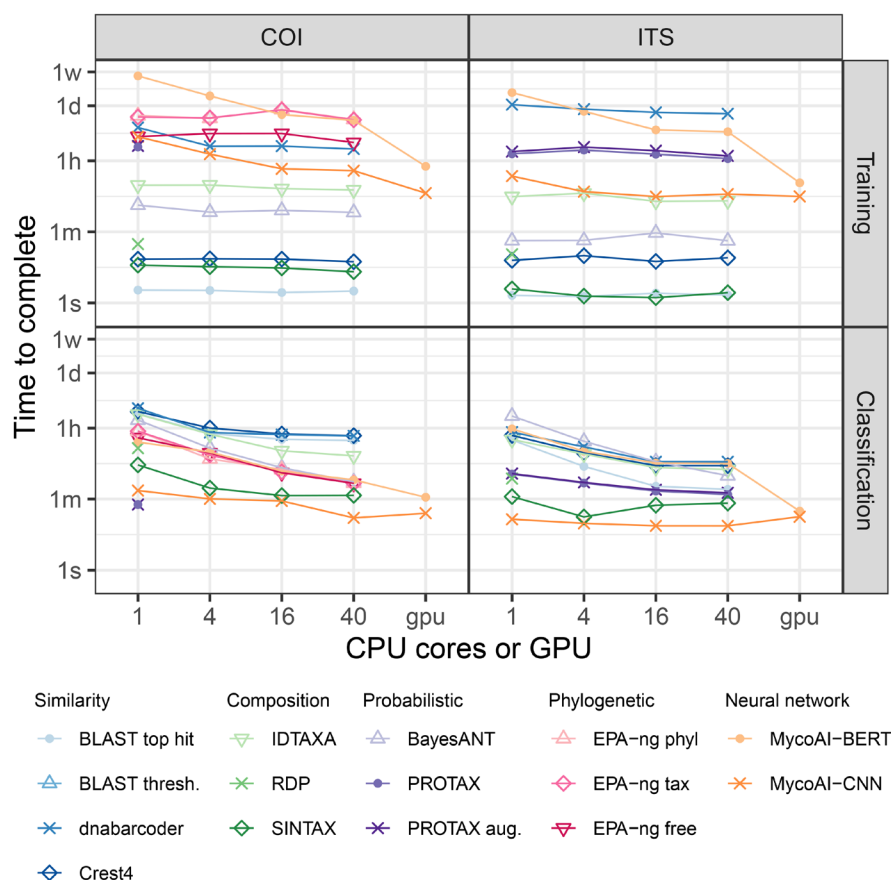


FIGURE 7 Runtime scaling for training and classification across methods and markers. Wall-clock time to complete (log scale; seconds to weeks) is shown for 1, 4, 16 and 40 CPU cores and GPU.

algorithms that abstain from making predictions when uncertain have a lower risk of misclassification. When dealing with novel taxa, this corresponds to the balance between under- and over-classification. Under-classification increases accuracy at higher ranks by withholding predictions when uncertainty is high, while over-classification not only increases coverage but also increases the chance of errors at lower ranks. Because algorithms vary in how they navigate this trade-off, their suitability depends on the objectives of a given study. For instance, biodiversity monitoring or community surveys may benefit from a high prediction coverage at the expense of accuracy, as detecting rare taxa is crucial (Mouillot et al., 2013; Soliveres et al., 2016). Conversely, applications such as network analysis may prioritize accuracy, since misclassifications can create spurious links between taxa (Cuff et al., 2022). To support informed decision-making, we argue that classification algorithms should provide well-calibrated confidence estimates. Clearly reporting classification uncertainties not only enhances scientific reliability but also enables researchers to incorporate these confidence estimates into downstream analyses, thereby refining subsequent inferences.

When reference databases are incomplete, algorithms should additionally distinguish between novelty and classification uncertainty, improving transparency and interpretability. Algorithms that rely on a single threshold to accept or reject classification must optimize thresholds to balance between different error types, often resulting in conservative cut-offs as observed in Figure 5. Treating confidence estimates below such thresholds as indicative of novelty can thus inflate estimates of unseen taxa. For instance, both BLAST threshold and dnabarcoder largely over-assigned DNA sequences to novel branches (Figure S7), highlighting that many 'under-classifications' reflect the stringency of the applied thresholds rather than true novelty. On the other hand, algorithms that explicitly account for the possibility of novel taxa and distinguish between novelty and classification uncertainty greatly enhance interpretability, enabling researchers to disentangle true biological signal from methodological conservatism and thereby draw more robust ecological and evolutionary inferences. Consequently, researchers should favour classification approaches that explicitly handle novelty when working in geographic regions or with taxonomic groups that are known to be underrepresented in reference databases. Accurate prediction of novelty not only improves ecological inference, but can serve to guide biodiversity discovery and database improvement. We argue that our approach of defining novelty within the existing taxonomic framework provides a more intuitive guidance than using for example sequence similarity thresholds: targeting sampling and sequencing campaigns of a given taxon is simpler than targeting taxa with a given sequence similarity to existing reference data—which may or may not correlate with either taxonomy or morphology.

4.2 | Different algorithm classes perform best on different genetic markers

Genetic markers vary in their suitability for taxonomic classification of different taxonomic groups, owing to lineage-specific rates of

evolution and variation in sequence composition (Hebert et al., 2003; Pentinsaari et al., 2016; Schoch et al., 2012). We evaluated classification algorithms on two widely used DNA barcodes chosen specifically for their contrasting characteristics: the COI barcode as a protein-coding gene with strong evolutionary constraints on the amino acid sequence and length, and ITS as a non-coding region of variable base composition and length.

While every method reached a high classification accuracy for observed species (i.e. the *observed everywhere* partition), we detected substantial differences in performance across genetic markers for novel taxa. For ITS, composition-based algorithms achieved the best results (e.g. Figure 3, green colours). This is likely due to the relatively smooth distribution of variability in the non-coding ITS, where positions close to each other show similar degrees of variability between taxa, allowing taxa at different degrees of relatedness to share a different fraction of *k*-mers. In contrast, the codon structure of COI results in conserved positions and highly variable positions interspersed at a small scale, reducing the potential for informative shared motifs. In this regard, the best performance for COI was achieved by phylogenetic placement (Figure 3, pink colours), which explicitly employs evolutionary models, including heterogeneous rates of evolution at different sites, to infer taxonomic relationships. Although phylogenetic placement is typically alignment-based, alternative phylogenetic placement algorithms relying on *k*-mer composition are available (Balaban et al., 2019), and could be an interesting future avenue to explore for taxonomic classification of ITS sequences. Overall, our results indicate that a better understanding of both sequence composition and evolutionary constraints can help researchers select the most appropriate algorithm for their specific application.

4.3 | Classification performance is not determined by computational resource use

We did not observe a general improvement in classification performance at the expense of higher computational costs. Indeed, the best performing algorithms for ITS included SINTAX, which had the lowest combined training and classification time, whereas computationally intensive phylogenetic placements were particularly rewarding for COI classifications. Although we assessed scalability in terms of available computational resources, we did not specifically examine how the models scale with the size of the reference database, which is an important practical consideration. Realistically, the size of the present databases might limit the use of resource-heavy algorithms. For phylogenetic placement, the computational efficiency may improve if the reference tree is built from a single sequence per species rather than the full set used in this study. It may even be necessary to preselect representative taxa for inclusion in the reference tree to make it computationally tractable.

4.4 | Limited impact of read length on classification performance

In many applications, shorter sub-regions of the DNA barcode are employed. Shorter reads often amplify more reliably than full-length barcodes (Meusnier et al., 2008) and are therefore preferable when working with degraded DNA. They can also be sequenced more cost-effectively on high-throughput platforms, making them attractive for large-scale barcoding studies. For most algorithms, we observed minimal differences in classification performance between short reads and full-length barcodes, suggesting that the regions targeted by the shorter reads captured most of the variation relevant for species classification. Previous studies have reported similar results for both COI (Yeo et al., 2020) and ITS (Badotti et al., 2017).

Notably, dnabarcoder failed to classify 99% of ITS short reads as belonging to the kingdom Fungi, resulting in near-zero accuracy, and MycoAI showed significantly reduced performance on short reads across genetic markers and data partitions. However, we trained all classifiers on the full-length barcodes and did not retrain them prior to classifying the short reads. The observed performance drop may therefore reflect sensitivity to mismatches between training and test sequence length, rather than an inherent inability of these algorithms to annotate short sequences.

4.5 | Barcode-based classifications under taxonomic uncertainty

To correctly classify novel taxa from DNA barcodes, the differences in the barcode sequence must reflect the true evolutionary relationships so that closely related groups have more similar barcodes than distant groups. For observed species, it is sufficient if interspecific variation of a genetic marker exceeds the intraspecific variation. For novel taxa, however, this criterion must be fulfilled also at higher taxonomic ranks, so that the barcode sequence composition is sufficient to determine the correct genus, family or even higher rank without an exact species match. This task is further complicated by the absence of universal divergence thresholds that reliably indicate the lowest shared taxonomic rank between two taxa (Edgar, 2018). In other words, even with the same genetic marker, a 3% divergence may correspond to taxa that share either a genus or a family. To support classification across a wide range of taxonomic ranks, it may therefore be beneficial to consider multiple genetic markers. For example, the 18S and 28S rRNA, which are immediately adjacent to the ITS, contain highly conserved regions which can be useful for taxonomic differentiation at higher ranks, as even small differences in these regions would indicate higher-level taxonomic divergence (Tedersoo, Mikryukov, et al., 2021). A further challenge is that there are incongruencies between currently accepted Linnean taxonomic classifications and molecular phylogenies, and taxonomic relationships are continually revised as new phylogenetic evidence corrects previous classifications (e.g. Chen et al., 2021; Ji et al., 2023; Johnston et al., 2024; Möckel et al., 2022). Beyond inconsistencies between reference databases, which differ in how

frequently they update their taxonomies, such incongruencies can undermine the consistency of barcode-based classification by violating the fundamental assumption that taxonomy accurately reflects evolutionary relatedness. These challenges highlight the appeal of methods that infer evolutionary relationships directly rather than relying solely on taxonomic labels. Phylogenetic placement is one such approach, integrating evolutionary signal into the classification process. However, its utility for taxonomic classification ultimately depends on how closely the reference taxonomy reflects true phylogenetic relationships.

Phylogenetic placement accuracy may also depend on the quality of the reference tree. Previous studies have shown that phylogenetic backbones can improve tree inference from DNA barcodes by supplementing their limited signal for deep evolutionary relationships (Liu et al., 2019; Talavera et al., 2022). To examine whether a more accurate phylogenetic representation of higher-level relationships translates into improved classification performance, we evaluated EPA-ng using three reference trees: one unconstrained, one constrained by taxonomy alone, and one incorporating phylogenetic relationships on the family level and above. We observed minimal differences in classification results between the two constrained reference trees, suggesting that constraining the tree to match taxonomy is generally sufficient for accurate classifications.

4.6 | Real-world representation and potential bias of the datasets

We used real-world geographic (FinBOL vs. GBOL) and methodological (Westerdijk vs. UNITE) splits to create partial taxonomic overlap between training and test sequences. For COI, the distribution shift between FinBOL and GBOL reflects both the difference in regional faunas and the distinct research priorities of their respective taxonomic communities, serving as an analogue of the regional mismatch between reference and query data when working in under-represented areas. For ITS, the shift instead stems mainly from methodological differences: the Westerdijk culture collection is restricted to fungi that can be grown in a laboratory, whereas UNITE includes a broader diversity of taxa collected from the environment. This, in turn, mirrors common real-world bias in reference data, which is disproportionately built from accessible, described, and readily sampled organisms (e.g., culturable strains and conspicuous macroscopic taxa).

However, neither the COI nor the ITS test data can capture the full range of characteristics of a truly real-world out-of-distribution scenario. For COI, although we capture two distinct geographic regions, they are relatively close, and larger biogeographic shifts (e.g., across continents or realms) could potentially introduce stronger changes in taxonomic composition, bringing in more distantly related lineages and thus more pronounced differences in the sequence patterns. The limited geographic range in our data could underestimate sequence differences within species (Bergsten et al., 2012), potentially inflating classifier performance. For ITS, curated reference datasets

remain biased toward culturable or well-studied taxa, whereas environmental samples can contain large proportions of poorly characterized diversity, including rare lineages, which may further challenge classification. For both COI and ITS, our data are curated to ensure that DNA sequences represent real species, rather than, for example, pseudogenes, PCR chimeras, or sequencing errors. Such artefacts are potentially common in non-curated environmental sequencing data, and classifiers may differ in how robustly they handle them. As a result, our performance estimates likely represent an upper bound on what can be expected in practice when working with raw environmental samples, given comparable coverage in the reference data. We encourage future benchmarks to explicitly evaluate classifier robustness to sequence quality degradation.

4.7 | Recommendations for algorithm selection

Our explicit assessment of classification performance on novel taxa provides a framework for guiding algorithm choice in studies targeting under-represented taxa. By evaluating both COI and ITS, two markers spanning a broad range of barcode characteristics, we also assess how robust these conclusions may be across barcodes. Although our results are specific to COI and ITS, and absolute accuracies are likely to be affected by differences in evolutionary rates between markers as well as database completeness and differing taxonomic practices, the observed trends may extend to other markers with similar properties. For example, among barcodes typically used for plants, we expect the same performance trends among algorithms for the functionally constrained, protein-coding chloroplast barcodes *matK* and *rbcl* as for COI (Guo et al., 2022), while trends for classification of the non-coding *trnH-psbA* barcode are likely comparable to those for ITS (Kress & Erickson, 2007), though this remains to be validated. Classification performance on barcodes with properties intermediate between COI and ITS, for example the large and small ribosomal DNA subunits, which contain strongly conserved regions interspersed with hypervariable regions (Staats et al., 2016), is likely intermediate between COI and ITS.

Overall, our results clearly show that the choice of taxonomic classifier is less critical when reference databases are complete, allowing researchers to prioritize computational efficiency. Nevertheless, even with comprehensive reference data, it remains important to choose an algorithm that provides confidence estimates to flag uncertain cases. For this purpose, our results highlight SINTAX as a fast, memory-efficient option that produces well-calibrated, albeit slightly conservative, classification confidence estimates.

When reference databases are incomplete, our results suggest that phylogenetic placement methods (e.g., EPA-ng) are preferable for COI barcodes, whereas composition-based methods (e.g. RDP-NBC, SINTAX and IDTAXA) perform best for ITS barcodes. Under incomplete reference databases, we further highlight the importance of selecting algorithms that explicitly distinguish between classification uncertainty and novel taxa, such as EPA-ng, PROTAX, and BayesANT. However, for ITS, no algorithm consistently achieved reliable identification of novel taxa across ranks. Consequently, the interpretation

of whether uncertain classifications indicate true novelty is left to the user, underscoring the importance of transparent criteria for assigning novelty and clear reporting of well-calibrated confidence estimates.

To reduce computational demands for large COI barcode datasets, we recommend a two-step strategy, as also suggested by Sundh et al. (2025). First, apply a fast and resource-efficient method such as SINTAX with a high confidence threshold to classify most observed species. Then, apply phylogenetic placement to the remaining uncertain cases to optimize classification. Because SINTAX confidence estimates for COI were consistently under-confident (Figure S9), a high confidence threshold would yield very few mis- or over-classifications.

Finally, our evaluations considered off-the-shelf approaches to taxonomic classification that leveraged default settings. In principle, one can further improve performance by properly tuning model-specific parameters to the training data and the specific marker, noting that some algorithms may be more sensitive to parameter choices than others (Hleap et al., 2021). Together, our results translate classifier benchmarking into practical guidance for algorithm choice under realistic barcode-classification scenarios, in which reference databases are incomplete, novel taxa are expected, and extensive parameter tuning is rarely feasible.

AUTHOR CONTRIBUTIONS

Johanna Orsholm: Co-first author. Formal analysis, investigation, visualization, writing—original draft; Alessandro Zito: Co-first author. Conceptualization, methodology, writing—review and editing; Panu Somervuo: Software, writing—review and editing; Jesse P. Harrison: Investigation, writing—review and editing; Markus Koskela: Investigation, writing—review and editing; Otso Ovaskainen: Investigation, writing—review and editing; Mariana P. Braga: Supervision, writing—review and editing; Nicolas Chazot: Investigation, supervision, writing—review and editing; Tomas Roslin: Supervision, writing—review and editing; Brendan Furneaux: Conceptualization, investigation, methodology, software, supervision, writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest in relation to this paper.

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DATA AVAILABILITY STATEMENT

Data and code are available via <https://doi.org/10.5281/zenodo.20704328> (Orsholm et al., 2026).

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REFERENCES

- Abarenkov, K. (2024). sh_matching_data_0_5_v9.zip: Supporting files for EOSC-Nordic service (SH matching analysis v2.0.0).
- Abarenkov, K., Nilsson, R. H., Larsson, K.-H., Taylor, A. S., May, T., Frøslev, T. G., Pawłowska, J., Lindahl, B., Pöldmaa, K., Truong, C., Vu, D., Hosoya, T., Niskanen, T., Piirmann, T., Ivanov, F., Zirk, A., Peterson, M., Cheeke, T., Ishigami, Y., ... Kõljalg, U. (2023). The UNITE database for molecular identification and taxonomic communication of fungi and other eukaryotes: Sequences, taxa and classifications reconsidered. *Nucleic Acids Research*, 52(D1), D791–D797.
- Abarenkov, K., Somervuo, P., Nilsson, R. H., Kirk, P. M., Huotari, T., Abrego, N., & Ovaskainen, O. (2018). Protax-fungi: A web-based tool for probabilistic taxonomic placement of fungal internal transcribed spacer sequences. *New Phytologist*, 220(2), 517–525.
- Abarenkov, K., Zirk, A., Piirmann, T., Pöhönen, R., Ivanov, F., Nilsson, R. H., & Kõljalg, U. (2025). UNITE general FASTA release for Fungi 2. Dataset.
- Altschul, S. F., Gish, W., Miller, W., Myers, E. W., & Lipman, D. J. (1990). Basic local alignment search tool. *Journal of Molecular Biology*, 215(3), 403–410.
- Badotti, F., de Oliveira, F. S., Garcia, C. F., Vaz, A. B. M., Fonseca, P. L. C., Nahum, L. A., Oliveira, G., & Góes-Neto, A. (2017). Effectiveness of its and sub-regions as DNA barcode markers for the identification of basidiomycota (fungi). *BMC Microbiology*, 17(1), 42.
- Balaban, M., Sarmashghi, S., & Mirarab, S. (2019). Apples: Scalable distance-based phylogenetic placement with or without alignments. *Systematic Biology*, 69(3), 566–578.
- Barbera, P., Kozlov, A. M., Czech, L., Morel, B., Darriba, D., Flouri, T., & Stamatakis, A. (2019). EPA-ng: Massively parallel evolutionary placement of genetic sequences. *Systematic Biology*, 68(2), 365–369.
- Bengtsson-Palme, J., Ryberg, M., Hartmann, M., Branco, S., Wang, Z., Godhe, A., De Wit, P., Sánchez-García, M., Ebersberger, I., de Sousa, F., Amend, A., Jumpponen, A., Unterseher, M., Kristiansson, E., Abarenkov, K., Bertrand, Y. J. K., Sanli, K., Eriksson, K. M., Vik, U., ... Nilsson, R. H. (2013). Improved software detection and extraction of its1 and its2 from ribosomal ITS sequences of fungi and other eukaryotes for analysis of environmental sequencing data. *Methods in Ecology and Evolution*, 4(10), 914–919.
- Bergsten, J., Bilton, D. T., Fujisawa, T., Elliott, M., Monaghan, M. T., Balke, M., Hendrich, L., Geijer, J., Herrmann, J., Foster, G. N., Ribera, I., Nilsson, A. N., Barraclough, T. G., & Vogler, A. P. (2012). The effect of geographical scale of sampling on DNA barcoding. *Systematic Biology*, 61(5), 851–869.
- BOLD Systems. (2024). Bold DNA barcode reference library. https://doi.org/10.5883/DP-BOLD_Public.29-Mar-2024
- Accessed via <https://boldsystems.org> on 2024-05-21.
- Bonato, K. O., Silva, P. C., Carvalho, F. R., & Malabarba, L. R. (2022). Trophic interactions of vampire catfishes (Siluriformes: Vandelliinae) revealed by metabarcoding analysis of stomach contents. *Freshwater Biology*, 67(3), 542–548.
- Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: Architecture and applications. *BMC Bioinformatics*, 10, 421.
- Chen, H., Lahey, Z., Talamas, E. J., Valerio, A. A., Popovici, O. A., Musetti, L., Klompen, H., Polaszek, A., Masner, L., Austin, A. D., & Johnson, N. F. (2021). An integrated phylogenetic reassessment of the parasitoid superfamily platygastroidea (Hymenoptera: Proctotrupomorpha) results in a revised familial classification. *Systematic Entomology*, 46(4), 1088–1113.
- Cuff, J. P., Windsor, F. M., Tercel, M. P., Kitson, J. J., & Evans, D. M. (2022). Overcoming the pitfalls of merging dietary metabarcoding into ecological networks. *Methods in Ecology and Evolution*, 13(3), 545–559.
- Czech, L., Barbera, P., & Stamatakis, A. (2020). Genesis and Gappa: Processing, analyzing and visualizing phylogenetic (placement) data. *Bioinformatics*, 36(10), 3263–3265.
- De Blasi, P., Favaro, S., Lijoi, A., Mena, R. H., Prünster, I., & Ruggiero, M. (2015). Are Gibbs-type priors the most natural generalization of the Dirichlet process? *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 37(2), 212–229.
- Deshpande, V., Wang, Q., Greenfield, P., Charleston, M., Porras-Alfaro, A., Kuske, C. R., Cole, J. R., Midgley, D. J., & Tran-Dinh, N. (2016). Fungal identification using a Bayesian classifier and the warcup training set of internal transcribed spacer sequences. *Mycologia*, 108(1), 1–5.
- Edgar, R. C. (2016). SINTAX: A simple non-Bayesian taxonomy classifier for 16S and ITS sequences. *bioRxiv*, 074161.
- Edgar, R. C. (2018). Accuracy of taxonomy prediction for 16S rRNA and fungal ITS sequences. *PeerJ*, 6, e4652.
- Elbrecht, V., Braukmann, T. W. A., Ivanova, N. V., Prosser, S. W. J., Hajibabaei, M., Wright, M., Zakharov, E. V., Hebert, P. D. N., & Steinke, D. (2019). Validation of COI metabarcoding primers for terrestrial arthropods. *PeerJ*, 7, e7745.
- Feibelman, T., Bayman, P., & Cibula, W. G. (1994). Length variation in the internal transcribed spacer of ribosomal DNA in chanterelles. *Mycological Research*, 98(6), 614–618.
- Folmer, O., Black, M., Hoeh, W., Lutz, R., & Vrijenhoek, R. (1994). DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. *Molecular Marine Biology and Biotechnology*, 3(5), 294–299.
- Furneaux, B., Bahram, M., Rosling, A., Yorou, N. S., & Ryberg, M. (2021). Long- and short-read metabarcoding technologies reveal similar

- spatiotemporal structures in fungal communities. *Molecular Ecology Resources*, 21(6), 1833–1849.
- German Barcode of Life Consortium. (2011). Zoological Research Museum Koenig—Leibniz Institute for Animal Biodiversity. Gbol webportal at <https://www.bolgermany.de>. [dataset] Version: 20170316.
- Gostel, M. R., & Kress, W. J. (2022). The expanding role of DNA barcodes: Indispensable tools for ecology, evolution, and conservation. *Diversity*, 14(3), 213.
- Guo, M., Yuan, C., Tao, L., Cai, Y., & Zhang, W. (2022). Life barcoded by DNA barcodes. *Conservation Genetics Resources*, 14(4), 351–365.
- Hebert, P. D. N., Cywinska, A., Ball, S. L., & deWaard, J. R. (2003). Biological identifications through DNA barcodes. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 270(1512), 313–321.
- Hebert, P. D. N., Stoeckle, M. Y., Zemlak, T. S., & Francis, C. M. (2004). Identification of birds through DNA barcodes. *PLoS Biology*, 2(10), e312.
- Hleap, J. S., Littlefair, J. E., Steinke, D., Hebert, P. D. N., & Cristescu, M. E. (2021). Assessment of current taxonomic assignment strategies for metabarcoding eukaryotes. *Molecular Ecology Resources*, 21(7), 2190–2203.
- Janzen, D. H., Hajibabaei, M., Burns, J. M., Hallwachs, W., Remigio, E., & Hebert, P. D. (2005). Wedding biodiversity inventory of a large and complex lepidoptera fauna with DNA barcoding. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 360(1462), 1835–1845.
- Ji, X., Sun, Y.-F., Wu, D.-M., Gao, N., & Cui, B.-K. (2023). An updated phylogenetic assessment and taxonomic revision of perenniporia sensu lato (polyporales, basidiomycota). *Journal of Fungi*, 9(2), 173.
- Johnston, N. P., Pape, T., Piwczyński, M., Wallman, J. F., Wiegmann, B. M., Cassel, B. K., Akbarzadeh, K., & Szpila, K. (2024). Anchored phylogenomics and revised classification of the miltogramminae (Diptera: Sarcophagidae). *Systematic Entomology*, 49(1), 138–155.
- Kalvari, I., Nawrocki, E. P., Ontiveros-Palacios, N., Argasinska, J., Lamkiewicz, K., Marz, M., Griffiths-Jones, S., Toffano-Nioche, C., Gautheret, D., Weinberg, Z., Rivas, E., Eddy, S. R., Finn, R. D., Bateman, A., & Petrov, A. I. (2021). Rfam 14: Expanded coverage of metagenomic, viral and microRNA families. *Nucleic Acids Research*, 49(D1), D192–D200.
- Khomich, M., Cox, F., Andrew, C. J., Andersen, T., Kausarud, H., & Davey, M. L. (2018). Coming up short: Identifying substrate and geographic biases in fungal sequence databases. *Fungal Ecology*, 36, 75–80.
- Kress, W. J., & Erickson, D. L. (2007). A two-locus global DNA barcode for land plants: The coding RBCL gene complements the non-coding trnH-psbA spacer region. *PLoS One*, 2(6), 1–10.
- Lanzén, A., Jørgensen, S. L., Huson, D. H., Gorfer, M., Grindhaug, S. H., Jonassen, I., Øvreås, L., & Urich, T. (2012). CREST—Classification resources for environmental sequence tags. *PLoS One*, 7(11), e49334.
- Leandro, C., Jay-Robert, P., & Pétilion, J. (2024). eDNA for monitoring and conserving terrestrial arthropods: Insights from a systematic map and barcode repositories assessments. *Insect Conservation and Diversity*, 17(4), 565–578.
- Lindahl, B. D., Nilsson, R. H., Tedersoo, L., Abarenkov, K., Carlsen, T., Kjoller, R., Kõljalg, U., Pennanen, T., Rosendahl, S., Stenlid, J., & Kausarud, H. (2013). Fungal community analysis by high-throughput sequencing of amplified markers—a user's guide. *New Phytologist*, 199(1), 288–299.
- Liu, J., Liu, J., Shan, Y.-X., Ge, X.-J., & Burgess, K. S. (2019). The use of DNA barcodes to estimate phylogenetic diversity in forest communities of southern China. *Ecology and Evolution*, 9(9), 5372–5379.
- Manter, D. K., & Vivanco, J. M. (2007). Use of the ITS primers, ITS1F and ITS4, to characterize fungal abundance and diversity in mixed-template samples by qPCR and length heterogeneity analysis. *Journal of Microbiological Methods*, 71(1), 7–14.
- Martin, M. (2011). Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal*, 17(1), 10–12.
- Meusnier, I., Singer, G. A., Landry, J.-F., Hickey, D. A., Hebert, P. D., & Hajibabaei, M. (2008). A universal DNA mini-barcode for biodiversity analysis. *BMC Genomics*, 9(1), 214.
- Möckel, L., Meusemann, K., Misof, B., Schwartze, V. U., De Fine Licht, H. H., Voigt, K., Stielow, B., de Hoog, S., Beutel, R. G., & Buellesbach, J. (2022). Phylogenetic revision and patterns of host specificity in the fungal subphylum entomophthoromycota. *Microorganisms*, 10(2), 256.
- Mouillot, D., Bellwood, D. R., Baraloto, C., Chave, J., Galzin, R., Harmelin-Vivien, M., Kulbicki, M., Lavergne, S., Lavorel, S., Mouquet, N., Paine, C. E. T., Renaud, J., & Thuiller, W. (2013). Rare species support vulnerable functions in high-diversity ecosystems. *PLoS Biology*, 11(5), e1001569.
- Murali, A., Bhargava, A., & Wright, E. S. (2018). IDTAXA: A novel approach for accurate taxonomic classification of microbiome sequences. *Microbiome*, 6(1), 140.
- Niskanen, T., Lücking, R., Dahlberg, A., Gaya, E., Suz, L. M., Mikryukov, V., Liimatainen, K., Druzhinina, I., Westrip, J. R., Mueller, G. M., Martins-Cunha, K., Kirk, P., Tedersoo, L., & Antonelli, A. (2023). Pushing the frontiers of biodiversity research: Unveiling the global diversity, distribution, and conservation of fungi. *Annual Review of Environment and Resources*, 48(1), 149–176.
- Orsholm, J., Zito, A., & Brendan, F. (2026). jorsholm/taxclass: v1.0.0 (Version v1.0.0) [Computer software]. Zenodo. <https://doi.org/10.5281/ZENODO.20704328>
- Pentinsaari, M., Salmela, H., Mutanen, M., & Roslin, T. (2016). Molecular evolution of a widely-adopted taxonomic marker (COI) across the animal tree of life. *Scientific Reports*, 6(1), 35275.
- Pitman, J. (1996). Some developments of the Blackwell-MacQueen urn scheme. In T. S. Ferguson, L. S. Shapley, & J. B. MacQueen (Eds.), *Statistics, probability and game theory. Papers in honor of David Blackwell, Volume 30 of IMS Lecture notes, Monograph Series* (pp. 245–267). Institute of Mathematical Statistics.
- Porter, T. M., & Hajibabaei, M. (2020). Putting COI metabarcoding in context: The utility of exact sequence variants (ESVs) in biodiversity analysis. *Frontiers in Ecology and Evolution*, 8, 248.
- Ranwez, V., Douzery, E. J. P., Cambon, C., Chantret, N., & Delsuc, F. (2018). MACSE v2: Toolkit for the alignment of coding sequences accounting for frameshifts and stop codons. *Molecular Biology and Evolution*, 35(10), 2582–2584.
- Ratnasingham, S., & Hebert, P. D. N. (2007). Bold: The barcode of life data system (<http://www.barcodinglife.org>). *Molecular Ecology Notes*, 7(3), 355–364.
- Ratnasingham, S., Wei, C., Chan, D., Agda, J., Agda, J., Ballesteros-Mejia, L., Boutou, H. A., El Bastami, Z. M., Ma, E., Manjunath, R., Rea, D., Ho, C., Telfer, A., McKeown, J., Rahulan, M., Steinke, C., Dorsheimer, J., Milton, M., & Hebert, P. D. N. (2024). BOLD v4: A centralized bioinformatics platform for DNA-based biodiversity data. Springer US.
- Riaz, T., Shehzad, W., Viari, A., Pompanon, F., Taberlet, P., & Coissac, E. (2011). ecoPrimers: Inference of new DNA barcode markers from whole genome sequence analysis. *Nucleic Acids Research*, 39(21), e145.
- Richardson, R. T., Bengtsson-Palme, J., & Johnson, R. M. (2017). Evaluating and optimizing the performance of software commonly used for the taxonomic classification of DNA metabarcoding sequence data. *Molecular Ecology Resources*, 17(4), 760–769.
- Rigon, T., Hsu, C.-L., & Dunson, D. B. (2025). A bayesian theory for estimation of biodiversity.
- Romeijn, L., Bernatavicius, A., & Vu, D. (2024). MycoAI: Fast and accurate taxonomic classification for fungal ITS sequences. *Molecular Ecology Resources*, 24(8), e14006.
- Roslin, T., Somervuo, P., Pentinsaari, M., Hebert, P. D. N., Agda, J., Ahlroth, P., Anttonen, P., Aspi, J., Blagoev, G., Blanco, S., Chan, D., Clayhills, T., deWaard, J., deWaard, S., Elliot, T., Elo, R., Haapala, S., Helve, E., Ilmonen, J., ... Mutanen, M. (2022). A molecular-based identification resource for the arthropods of Finland. *Molecular Ecology Resources*, 22(2), 803–822.

- Schoch, C. L., Seifert, K. A., Huhndorf, S., Robert, V., Spouge, J. L., Levesque, C. A., Chen, W., & Consortium, F. B. (2012). Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for fungi. *Proceedings of the National Academy of Sciences of the United States of America*, 109(16), 6241–6246.
- Shokralla, S., Gibson, J. F., Nikbakht, H., Janzen, D. H., Hallwachs, W., & Hajibabaei, M. (2014). Next-generation DNA barcoding: Using next-generation sequencing to enhance and accelerate DNA barcode capture from single specimens. *Molecular Ecology Resources*, 14(5), 892–901.
- Sickel, W., Ankenbrand, M. J., Grimmer, G., Holzschuh, A., Härtel, S., Lanzen, J., Steffan-Dewenter, I., & Keller, A. (2015). Increased efficiency in identifying mixed pollen samples by meta-barcoding with a dual-indexing approach. *BMC Ecology*, 15(1), 20.
- Soliveres, S., Manning, P., Prati, D., Gossner, M. M., Alt, F., Arndt, H., Baumgartner, V., Binkenstein, J., Birkhofer, K., Blaser, S., Blüthgen, N., Boch, S., Böhm, S., Börschig, C., Buscot, F., Diekötter, T., Heinze, J., Hölzel, N., Jung, K., ... Allan, E. (2016). Locally rare species influence grassland ecosystem multifunctionality. *Philosophical Transactions of the Royal Society, B: Biological Sciences*, 371(1694), 20150269.
- Somervuo, P., Koskela, S., Pennanen, J., Henrik Nilsson, R., & Ovaskainen, O. (2016). Unbiased probabilistic taxonomic classification for DNA barcoding. *Bioinformatics*, 32(19), 2920–2927.
- Somervuo, P., Yu, D. W., Xu, C. C., Ji, Y., Hultman, J., Wirta, H., & Ovaskainen, O. (2017). Quantifying uncertainty of taxonomic placement in DNA barcoding and metabarcoding. *Methods in Ecology and Evolution*, 8(4), 398–407.
- Staats, M., Arulandhu, A. J., Gravendeel, B., Holst-Jensen, A., Scholtens, I., Peelen, T., Prins, T. W., & Kok, E. (2016). Advances in DNA metabarcoding for food and wildlife forensic species identification. *Analytical and Bioanalytical Chemistry*, 408(17), 4615–4630.
- Stork, N. E. (2018). How many species of insects and other terrestrial arthropods are there on earth? *Annual Review of Entomology*, 63(1), 31–45.
- Sundh, J., Granqvist, E., Iwaszkiewicz-Eggebrecht, E., Manoharan, L., van Dijk, L. J., Goodsell, R., Godeiro, N. N., Bellini, B. C., Orsholm, J., Łukasik, P., Miraldo, A., Roslin, T., Tack, A. J. M., Andersson, A. F., & Ronquist, F. (2025). Hap: High-accuracy pipeline for processing deep metabarcoding data. *PLoS Computational Biology*, 21(11), e1013558.
- Taberlet, P., Coissac, E., Pompanon, F., Brochmann, C., & Willerslev, E. (2012). Towards next-generation biodiversity assessment using DNA metabarcoding. *Molecular Ecology*, 21(8), 2045–2050.
- Talavera, G., Lukhtanov, V., Pierce, N. E., & Vila, R. (2022). DNA barcodes combined with multilocus data of representative taxa can generate reliable higher-level phylogenies. *Systematic Biology*, 71(2), 382–395.
- Tedersoo, L., Albersen, M., Anslan, S., & Callahan, B. (2021). Perspectives and benefits of high-throughput long-read sequencing in microbial ecology. *Applied and Environmental Microbiology*, 87(17), e00626–21.
- Tedersoo, L., Bahram, M., Zinger, L., Nilsson, R. H., Kennedy, P. G., Yang, T., Anslan, S., & Mikryukov, V. (2022). Best practices in metabarcoding of fungi: From experimental design to results. *Molecular Ecology*, 31(10), 2769–2795.
- Tedersoo, L., Mikryukov, V., Anslan, S., Bahram, M., Khalid, A. N., Corrales, A., Agan, A., Vasco-Palacios, A.-M., Saitta, A., Antonelli, A., Rinaldi, A. C., Verbeken, A., Sulistyo, B. P., Tamgnoue, B., Furneaux, B., Ritter, C. D., Nyamukondiwa, C., Sharp, C., ... Abarenkov, K. (2021). The global soil mycobiome consortium dataset for boosting fungal diversity research. *Fungal Diversity*, 111(1), 573–588.
- Van Klink, R., August, T., Bas, Y., Bodesheim, P., Bonn, A., Fossøy, F., Høye, T. T., Jongejans, E., Menz, M. H., Miraldo, A., Roslin, T., Roy, H. E., Ruczyński, I., Schigel, D., Schäffler, L., Sheard, J. K., Svenningsen, C., Tschan, G. F., Wäldchen, J., ... Bowler, D. E. (2022). Emerging technologies revolutionise insect ecology and monitoring. *Trends in Ecology & Evolution*, 37(10), 872–885.
- Vass, M., Eriksson, K., Carlsson-Graner, U., Wikner, J., & Andersson, A. (2022). Co-occurrences enhance our understanding of aquatic fungal metacommunity assembly and reveal potential host-parasite interactions. *FEMS Microbiology Ecology*, 98(11), fiac120.
- Vu, D., Groenewald, M., De Vries, M., Gehrman, T., Stielow, B., Eberhardt, U., Al-Hatmi, A., Groenewald, J. Z., Cardinali, G., Houbraken, J., Boekhout, T., Crous, P. W., Robert, V., & Verkley, G. J. M. (2019). Large-scale generation and analysis of filamentous fungal DNA barcodes boosts coverage for kingdom fungi and reveals thresholds for fungal species and higher taxon delimitation. *Studies in Mycology*, 92(1), 135–154.
- Vu, D., Groenewald, M., Szöke, S., Cardinali, G., Eberhardt, U., Stielow, B., De Vries, M., Verkley, G., Crous, P., Boekhout, T., & Robert, V. (2016). DNA barcoding analysis of more than 9 000 yeast isolates contributes to quantitative thresholds for yeast species and genera delimitation. *Studies in Mycology*, 85(1), 91–105.
- Vu, D., Nilsson, R. H., & Verkley, G. J. M. (2022). Dnabarcoder: An open-source software package for analysing and predicting DNA sequence similarity cutoffs for fungal sequence identification. *Molecular Ecology Resources*, 22(7), 2793–2809.
- Wang, Q., Garrity, G. M., Tiedje, J. M., & Cole, J. R. (2007). Naïve Bayesian classifier for rapid assignment of rRNA sequences into the new bacterial taxonomy. *Applied and Environmental Microbiology*, 73(16), 5261–5267.
- Winand, R., D'hooge, E., Van Uffelen, A., Bogaerts, B., Van Braekel, J., Hoffman, S., Roosens, N. H. C. J., Becker, P., De Keersmaecker, S. C. J., & Vanneste, K. (2025). Investigating fungal diversity through metabarcoding for environmental samples: Assessment of ITS1 and ITS2 Illumina sequencing using multiple defined mock communities with different classification methods and reference databases. *BMC Genomics*, 26(1), 729.
- Yeo, D., Srivathsan, A., & Meier, R. (2020). Longer is not always better: Optimizing barcode length for large-scale species discovery and identification. *Systematic Biology*, 69(5), 999–1015.
- Zito, A., Rigon, T., & Dunson, D. B. (2023). Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa. *Methods in Ecology and Evolution*, 14(2), 529–542.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Illustration of how the constraint tree was edited to ensure monophyletic families. In cases where a family was not monophyletic in the original tree, the branch for that family was placed as child of the last common ancestor of that family in the original tree.

Figure S2. Accuracy and marginal recall of classification algorithms across taxonomic ranks under the FP setting, where predictions were enforced for all test sequences. This corresponds to main **Figure 3**, here showing results expanded across algorithm categories. (a) shows accuracy, defined as the proportion of all taxonomic classifications that were correct for sequences in the *observed everywhere* and *novel anywhere* partitions. (b) shows marginal recall, calculated as the proportion of correct predictions relative to the total number of sequences belonging to either *observed taxa* or *novel taxa* at that rank.

Figure S3. Accuracy and marginal recall of classification algorithms across taxonomic ranks under the MP setting, that is where predictions were enforced for all query sequences. (a) shows accuracy, that is the proportion of all taxonomic classifications that were correct for

sequences in the *observed everywhere* and *novel anywhere* partitions. (b) shows marginal recall, that is the proportion of correct predictions relative to the total number of sequences belonging to that respective taxon set and rank for *observed taxa* or *novel taxa*.

Figure S4. Conditional recall for taxonomic predictions of novel taxa. For each rank, it shows the proportion of correct predictions given that the rank above was classified correctly. Each algorithm is represented by a different shape and colour, with hue indicating algorithm group: sequence similarity (blue), composition (green), probabilistic (purple), phylogenetic (pink) and neural network (orange). Numbers display the count of correct predictions on the rank above for each algorithm and rank, that is the number of predictions included in the denominator.

Figure S5. Rates of mis-, over- and under-classification of COI and ITS sequences across classification algorithms and taxonomic ranks. Error rates are shown under the MP setting, allowing missing predictions and implementing confidence thresholds for algorithms that recommend them. This corresponds to main Figure 4, here showing results expanded across algorithm categories.

Figure S6. Classification accuracy as a function of prediction coverage for different algorithms. The x-axis shows the proportion of sequences classified and the y-axis shows the proportion of correct predictions. Continuous lines indicate algorithms with confidence estimates, where varying the threshold yields a trade-off between coverage and accuracy. Single points represent algorithms without confidence estimates, corresponding to a fixed trade-off. We allowed missing predictions but did not implement confidence thresholds for algorithms that recommend them. However, recommended thresholds are indicated by points along the continuous line. This corresponds to main Figure 5, here showing results expanded across algorithm categories.

Figure S7. Cumulative number of taxa predicted as novel across ranks and classification algorithms. The true number of novel taxa is shown with a black dashed line. Only algorithms which explicitly predict novel taxa or produce missing predictions are included in the plot.

Figure S8. Change in classification accuracy when using nucleotide sequences versus amino acid translation of the COI barcode, shown in percentage points. Line of zero difference shown with a black dashed line. Negative values indicate lower accuracy of amino acid sequences, and vice versa. The top row shows result for the *observed everywhere* partition, and the bottom row shows result for the *novel anywhere* partition.

Figure S9. Calibration of taxonomic predictions across ranks for COI. The x-axis shows the cumulative probability of the prediction with the highest probability or confidence score and the y-axis shows the cumulative proportion of correct predictions. Models are well calibrated if the line follows the identity line, here shown in grey. The y coordinate of the point at the end of the line displays the average accuracy. Calibration is shown for the partitions *observed everywhere* (COI_obs) and *novel anywhere* (COI_novel). Calibration results are under the FP setting, that is where predictions are enforced for all query sequences.

Figure S10. Calibration of taxonomic predictions across ranks for ITS. The x-axis shows the cumulative probability of the prediction with the highest probability or confidence score and the y-axis shows the cumulative proportion of correct predictions. Models are well calibrated if the line follows the identity line, here shown in grey. The y coordinate of the point at the end of the line displays the average accuracy. Calibration is shown for the partitions *observed everywhere* (ITS_obs) and *novel anywhere* (ITS_novel). Calibration results are under the FP setting, that is where predictions are enforced for all query sequences.

Table S1. Additional algorithms for taxonomic classification and why they were not included in our evaluation.

Table S2. Origin of the backbone trees used to build the phylogenetically constrained reference tree.

Table S3. Computational resource use of different classification algorithms. Initialization/training and classification phases are shown separately for COI and ITS sequences. Reported metrics include wall-clock time, average CPU usage, peak memory consumption, and model size on disk. Values in parentheses are for classification of short sequences representative of metabarcoding data. Results are provided for both single-core and multi-core CPU runs, as well as GPU runs where available.

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